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Motor Neuron Diseases

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SUMMARY PEARLS

- Motor neuron diseases (MNDs) are a group of neuromuscular disorders that affect the anterior horn cells or the cortical motor neurons, or both.
- Clinical features of MNDs may include both upper motor neuron features (spasticity, increased tone, or brisk reflexes) and lower motor neuron features (weakness, atrophy, and fasciculations).
- An important characteristic of most MNDs is the absence of overt sensory symptoms such as pain, numbness, and tingling.
- MNDs can be inherited or acquired, manifesting in either childhood or adulthood.
- Amyotrophic lateral sclerosis (ALS) is characterized by its unique involvement of both upper and lower motor neurons, resulting in distinctive clinical manifestations.
- Spinal muscular atrophy (SMA), once the leading genetic cause of infant mortality, now has three available disease-modifying treatments that target the genetic cause of the disease.
- Poliomyelitis, a viral-mediated MND, has been largely eradicated globally, though rare cases still occur because of wild-type and vaccine-derived strains as well as other infections, such as West Nile virus.
- Less common forms of MNDs include Hirayama disease, distal SMA, and paraneoplastic MND.
- ALS is universally fatal, yet various symptomatic and disease-modifying treatments, along with rehabilitation measures, can significantly prolong and improve the quality of life for affected patients.
- Rehabilitation measures for MNDs are essential to enhancing function, extending survival, and improving the quality of life for patients with MNDs.

Introduction

A floppy infant, painless foot drop, and a paralyzed child are potential manifestations of motor neuron disease (MND). Few diseases in medicine cause as much concern as presentations of spinal muscular atrophy (SMA), amyotrophic lateral sclerosis (ALS), and poliomyelitis. Few diseases as rare as ALS are a household name, yet Lou Gehrig disease (synonymous with ALS or simply MND) is widely known even by those who never saw the “Iron Horse” play baseball or know the full story of the great man after whom the disease is named. MNDs include a heterogeneous group of disorders associated with dysfunction and degeneration of motor neurons, alone or in combination with other neuron pools. The selective and relentless degeneration or dysfunction of motor neurons, a common thread of most MNDs, invokes a remarkable yet rational degree of fear in patients and providers alike. This can only recently be mixed with hope, as new breakthroughs are changing the grim prognosis of some forms of MND. The objective of this chapter is to provide an overview of the spectrum of MNDs and outline the general approach to evaluation, diagnosis, and management of MNDs.

General Features of Motor Neuron Disease

Classification

MNDs can be broadly grouped into inherited or acquired causes. The main acquired or sporadic causes include poliomyelitis, ALS,

and ALS variants. Rarer acquired forms include immune-mediated, endocrine, traumatic, nonpoliomyelitis infections and paraneoplastic etiologies. The most common inherited causes of MND include SMA, familial ALS, and X-linked bulbospinal muscular atrophy (Kennedy disease [KD]). MNDs can also be divided into groups based on selectivity for loss of upper motor neurons (UMNs; corticospinal and corticobulbar tracts) or lower motor neurons (LMNs; spinal anterior horn cells and cranial nerve LMNs) (Box 41.1).

Assessment

History

A thorough history and physical examination are vital when assessing a patient with a possible MND. Patients with MNDs usually present with painless weakness. The initial presentation is focal in Hirayama disease (HD) and can be in ALS and poliomyelitis. However, less prominent generalized weakness is also often present, but it is only found on a directed history. Associated features, such as prominent sensory symptoms or complaints attributable to sphincter or autonomic dysfunction, are unusual in pure motor neuron processes and should prompt consideration of an alternative process. The history should begin by focusing on when the patient (or family member) first noticed symptoms along with the distribution and characteristics of these symptoms. Establishing the rate of progression is helpful. A very slowly progressive course may suggest an inherited disorder. In this case, a

• BOX 41.1 Classification of Motor Neuron Diseases

Mixed Upper and Lower Motor Neurons

Amyotrophic lateral sclerosis (ALS)
ALS-plus

Upper Motor Neuron

Primary lateral sclerosis

Lower Motor Neuron

ALS-related variants with predominant lower motor neuron loss

- Progressive muscular atrophy
- Progressive bulbar palsy
- Brachial amyotrophic diplegia (flail arm syndrome)
- Leg amyotrophic diplegia (flail leg syndrome)

Acute poliomyelitis and postpolio syndrome

Spinal muscular atrophy (SMA)

- Proximal
- Distal (hereditary motor neuropathy)
- Scapuloperoneal SMA (Davidenkow syndrome)
- Kennedy disease (X-linked spinobulbar muscular atrophy)
- Hirayama disease (monomelic SMA)

Less Common or Less Well-Defined Syndromes

- Paraneoplastic
- Nonpolio-related infections
- Electrical injury
- Hereditary metabolic disorders
- Hereditary bulbar syndromes

patient may be uncertain about the duration and rate of progression. Asking about difficulty with specific tasks is also important. A thorough family history is essential to assess for any clues of an underlying hereditary process, and sometimes examination of family members is necessary to identify similar but unrecognized clinical features. When interviewing a patient with suspected MND, it is recommended that a history covering at least three generations is obtained and that the presence of not only symptoms of MND but also frontotemporal dementia (FTD), other dementias, parkinsonism, psychiatric disease, and suicidal tendencies is sought because they may be varied manifestations of the same mutation if ALS is suspected.¹⁶⁵ An autosomal dominant pattern is especially important because the majority of identified genetic causes of ALS are transmitted in this manner.¹⁶⁵

Respiratory or bulbar involvement can be subtle and may take some probing to identify; features such as orthopnea or morning headaches may suggest diaphragm paralysis or nocturnal hypoventilation, respectively. The past medical history and review of systems can identify any systemic processes that may be related to MND. The functional and social history will be critical in prescribing orthotics, assistive devices, and therapy as well as eventually modifying the patient's living situation as the disease progresses. Physicians should expect to become familiar with family support and living conditions to anticipate transitional challenges with disease progression.

Physical Exam

A detailed physical examination can help distinguish between different MNDs and tailor their diagnostic and management approaches. Examination should detail the presence of muscle atrophy, weakness, fasciculations and alterations of muscle tone, reflexes, bulbar function, gait, and coordination. Muscle atrophy,

weakness, and fasciculations are the primary features of LMN dysfunction that should be sought. Fasciculations may be obvious on general observation or may require prolonged inspection of various muscle groups. Tongue fasciculations are the most readily identified evidence of LMN injury in the bulbar muscles. Fasciculations near the surface of the muscle are visible on exam, but fasciculations occurring deep within the muscle may only be seen with needle electromyography or with muscle ultrasound.¹³⁴ The implications of fasciculations depend almost entirely on the clinical context. In isolation, fasciculations are not pathognomonic for MND and can be seen in healthy individuals as well as numerous other conditions of the peripheral nervous system. Fasciculations should be interpreted in the context of other concomitant clinical and electrodiagnostic findings of active and chronic denervation. Findings of fasciculations in a patient with weakness and evidence of denervation on electromyography (EMG) are strongly suggestive of MND, but when fasciculations are found in isolation, they are nonspecific and may not be pathologic.

Laboratory Studies

The laboratory approach to the patient with possible MND should be guided by features on history and clinical examination and is generally used to help exclude alternate diagnoses. Creatine kinase (CK) levels are often modestly elevated in MND and may confuse providers when evaluating a patient with painless weakness. Muscle biopsies are sometimes ordered unnecessarily because of the confusion this creates. Muscle biopsy is not normally necessary in patients with suspected MND unless the presentation is atypical or myopathy is suspected. Neurofilament light chain (NFL) is a more recently available laboratory test that becomes elevated in the blood or cerebrospinal fluid (CSF) when axonal damage occurs in the central or peripheral nervous system. NFL elevation, similar to how elevated CK levels indicate muscle damage in myopathies, can be used to detect, identify, and roughly quantify the degree of axon loss. Studies have found that NFL levels in CSF and blood samples can differentiate ALS patients from other neurologic conditions and healthy individuals.¹⁹⁶ The potential of NFL levels as markers for monitoring disease progression and their correlation with disease severity is still unclear. More research is needed to confirm these associations and establish their reliability.

Genetic testing can, in some cases, confirm the diagnosis of hereditary MNDs, and other testing may not be warranted. Such examples may include testing for homozygous deletion or mutation of the *SMN1* gene for proximal SMA or testing for cytosine-adenine-guanine (CAG) trinucleotide repeat number for possible KD. As genetic testing technologies and our understanding of the genetic causes of MND have improved, there is less reliance on an a priori approach to testing. Instead, multigene panels or exome or genome sequencing are now commonly used.

Imaging

Imaging studies may not be required in the evaluation of patients with suspected MND but are useful and sometimes of vital importance to help exclude mimicking or confounding diagnoses causing UMN or LMN involvement.

Electrodiagnosis

Electrodiagnostic testing is the primary diagnostic modality to confirm loss or dysfunction of the peripheral nervous system. Electrodiagnostic studies in patients with suspected MND are designed to identify LMN involvement and to exclude other

mimicking disorders of the peripheral nervous system. Needle EMG is the most pertinent aspect of the electrodiagnostic examination for identification of LMN loss, but nerve conduction studies (NCS) are important to help exclude other diagnostic possibilities within the peripheral nervous system. Usually, needle EMG features of LMN involvement, such as fibrillation potentials, fasciculation potentials, or neurogenic changes in motor unit action potential (MUAP) characteristics, are more readily apparent than clinical features of LMN involvement. Motor NCS may remain normal until there is sufficient motor axonal loss to result in diminished compound muscle action potential amplitudes.²⁷ In general, parameters of action potential propagation, including distal latencies and conduction velocities, will be relatively normal. Incidental findings of an asymptomatic or minimally symptomatic neuropathy with symptoms occurring over a different time frame should not distract from the overall pattern of a significant, primarily motor process. Some MNDs are associated with characteristic concomitant sensory involvement, although this is less severe than the findings of motor axonal loss. One important example is KD, which is usually associated with non-length-dependent sensory amplitude loss consistent with a coexistent sensory ganglionopathy. Other parameters of nerve conduction, including the latencies of F-waves and H-reflexes, are relatively preserved in proportion to the motor axonal loss.

Multifocal motor neuropathy (MMN) is one particular entity that deserves attention during the electrodiagnostic evaluation because it can closely mimic MND. MMN is an autoimmune disorder of the peripheral nerve associated with multifocal conduction block. Focal areas of conduction block in motor studies (but not sensory studies) outside of areas prone to entrapment can be seen in MMN. The motor abnormalities in MMN are typically more prominent in the upper limbs vs. the lower limbs and in distal limb muscles vs. proximal limb muscles. Multiple motor studies with multiple stimulation sites may be needed in some patients with suspected ALS to exclude subtle areas of conduction block that can be seen in MMN. Importantly, some patients with otherwise typical MMN will not have features of conduction block but still have favorable response to immunomodulatory treatment. Another distinctive feature that is typically evident on electrodiagnostic assessment of MMN is neurogenic involvement in the distribution of peripheral nerves rather than in the myotomal distribution that is otherwise typical of MNDs.

Neuromuscular junction (NMJ) disorders also result in clinical phenotypes of pure motor deficits and should be high in the differential diagnosis of a patient with suspected MND. Clinical features of prominent ocular, bulbar, and proximal limb weakness with fatigability are characteristic of a typical primary NMJ transmission defect. Myasthenia gravis can most closely mimic MND with bulbar and LMN predominant involvement, such as progressive bulbar palsy (PBP). Electrodiagnostic studies to identify failure of NMJ transmission, including repetitive nerve stimulation (RNS) or single-fiber EMG (SFEMG), should be considered, particularly in cases with evidence of fatigability on examination. Unfortunately, findings of NMJ transmission failure are not specific to primary NMJ disorders. Abnormal decrement may be seen on RNS studies in 20% of patients with ALS.⁸⁸ Similarly, SFEMG abnormalities of jitter and blocking can be seen in muscles that are normal on standard needle EMG in patients with MNDs, but such findings are usually associated with increased fiber density. The RNS and SFEMG abnormalities seen in MND are likely related to the inability of failing motor neurons to maintain sufficient synaptic output and the reduced safety factor at

newly formed, immature synapses after sprouting and collateral reinnervation. This can make distinguishing primary NMJ disorders from MND more difficult. However, other findings on needle EMG can further distinguish the two disease entities; for instance, neurogenic changes in MUAP size or recruitment are present in MND but absent in NMJ disorders. Fibrillations are typically present in MND but can also occur in some NMJ disorders, including botulism, and rarely other disorders.

Other less common but helpful electrodiagnostic techniques are sometimes utilized in the diagnosis of MND for some patients. Macro EMG and motor unit number estimation (MUNE) are specialized techniques that can investigate motor unit function. Although usually utilized in the research setting, MUNE has shown promise as an outcome measure in preclinical and clinical trials of different types of MND, including SMA, ALS, and KD.

Motor Neuron Disease Treatment

General

Treatment of MND varies related to factors such as type of MND, disease duration, and severity. The majority of MNDs lack effective disease-modifying medical interventions (e.g. medications) to halt or counteract disease progression. Accordingly, multidisciplinary supportive care is essential to address disease impact and improve or maintain function. Treatments specific for individual MND will be discussed later with those individual disorders, as well as rehabilitation approaches and symptomatic and medical treatment for MNDs in general. [Box 41.2](#) summarizes ALS and SMA treatments.

Disease-Modifying Medications

The majority of MNDs have no or limited treatment, and historically there have been no treatments available with a large disease-modifying effect for MNDs. Yet, at an ever-increasing rate, newer therapies are being developed for MNDs. Currently, there are multiple disease-modifying therapies for both ALS and SMA. Timely implementation of treatments to prevent (or slow) motor neuron loss is essential, and treatment should begin without delay.

Specific Diseases

ALS and ALS Variants

Sporadic and Familial ALS

ALS has an incidence of 1.4/100,000 and typically begins in the sixth to seventh decade of life.¹⁴⁴ Males are affected somewhat more than females, with a ratio of 1.6:1.¹³⁷ An important and distinctive feature of ALS is the degeneration of both the UMNs and LMNs, with relative sparing of other neurons. Although the understanding of ALS is ever expanding, the pathogenic mechanisms remain incompletely defined. There are hereditary (familial) forms (fALS) and sporadic forms (sALS) of ALS, and these are indistinguishable on clinical presentation and evaluation.

Roughly 10% to 20% of patients with apparent sALS due to lacking family history are found to have a known genetic mutation associated with fALS.¹⁹ Superoxide dismutase 1 (*SOD1*) mutations were the first to be identified as a causative etiology of fALS.²¹ Mutations in *SOD1* represent approximately 10% of fALS and are seen in approximately 1% of sALS. Even within the spectrum of *SOD1*-related ALS, there is considerable phenotypic variability. For instance, the *D90A* mutation of *SOD1* portends an indolent course with ventilatory failure often occurring 10 years

• BOX 41.2 Key Recommendations in Specific Motor Neuron Diseases

Summary of Key Recommendations in the Treatment of Amyotrophic Lateral Sclerosis

Exercise

- Some form of therapeutic exercise should be offered to every patient with amyotrophic lateral sclerosis (ALS)
- Range of motion and stretching exercises should be offered to every patient
- A gentle aerobic exercise program should be offered to ambulatory patients
- A resistance exercise program of mild-moderate resistance exercise for 15 minutes should be offered
- A resistance exercise program of mild-moderate resistance exercise for 15 minutes twice daily should be offered to all high-functioning patients
- Patients should avoid:
 - Exercising to fatigue
 - Exercising to dyspnea
 - Eccentric exercise
- Patients should scale back their program if they have:
 - Muscle cramping
 - Excessive overuse weakness afterward

Nutrition

- Monitor weight every 3 months
- Percutaneous endoscopic gastrostomy (PEG) or radiologically inserted gastrostomy (RIG) tube
- Indications:
 - Weight loss >10% of body weight
 - Dehydration
 - Meals limited by dysphagia and choking
 - Not aspiration pneumonia
- Ensure all patients have accurate information regarding risks and survival advantages of PEG or RIG
- Refer willing patients for PEG or RIG before forced vital capacity (FVC) drops to <50% predicted
- Refer willing patients with FVC between 50% and 30% for RIG

Medications

Riluzole

- Should be offered to every patient with ALS
- Dosage is 50 mg twice daily
- Treatment should be initiated as soon as possible
- Physicians should ensure patients have accurate information regarding riluzole
- Patients on riluzole should have liver function tests monitored

Edaravone

- Has been shown in one trial to help a select group of patients with ALS (Al-chalabi)^{5a,149a}
- Was shown to be of no significant benefit to a less specific group of patients with ALS (Al-chalabi)
- Is generally well tolerated, with no known drug-drug interactions (Oskarsson 2018)
- Has not been shown to improve quality of life or survival (Al-chalabi)
- Is very expensive and requires frequent infusions

Tofersen

- For patients with SOD1-ALS
- Monthly intrathecal injections
- Alternative Medications
- Patients should be advised that no other medication, nutritional supplement, or alternative medication has been shown to help patients with ALS

Multidisciplinary Clinics

- Extend survival and should be pursued when feasible

Mood Disorders

- Are common and should be treated to improve quality of life

Spasticity

- Intrathecal baclofen has been shown to work, but oral baclofen has not
- Botulinum toxin can be used but can lead to generalized weakness

Respiratory

- Respiratory measurements (usually FVC and maximal inspiratory pressure [MIP]) should be measured at every visit regardless of symptoms

- Noninvasive ventilation (NIV)
 - Extends survival, slows respiratory decline, and improves quality of life
 - Commonly begun when FVC <50% predicted and MIP ≤60 cm H₂O or a PaCO₂ of ≥45 mm Hg, although early NIV prolongs life
- Supplemental O₂ suppresses respiratory drive and can worsen symptoms and lead to respiratory arrest and should only be used for palliation of symptomatic hypoxia
- A mechanical insufflator-exsufflator improves secretion clearance and can be used in patients with a peak cough flow <300 to 350 L/min

Communication

- Is a major determinant of quality of life and should be optimized
- Patients should be routinely monitored by a speech-language pathologist

Palliative Care

- Discuss the patient's wishes and involve a palliative care team as early as possible
- Patient's distressing symptoms are often undertreated, and benzodiazepines and opioids titrated to clinical effect will almost never cause life-threatening respiratory depression

Summary of Key Recommendations in the Treatment of SMA

Exercise

- Tailored physical therapy programs should be provided to maintain mobility and prevent joint contractures
- Range of motion exercises and stretching should be included
- Moderate aerobic activities should be encouraged as tolerated

Nutrition

- Regular nutritional assessments are crucial to manage feeding difficulties and ensure adequate caloric intake
- Gastrostomy tube feeding should be considered for patients with severe swallowing difficulties or significant weight loss
- Consult a dietitian for individualized dietary plans

Medications

Nusinersen

- Approved for all types of SMA
- Administered via intrathecal injection
- Shown to improve motor function and survival

Onasemnogene abeparvovec

- Approved for patients age <2 years
- A single-dose intravenous gene therapy
- Aims to replace the defective *SMN1* gene

Risdiplam

- Approved for patients age ≥2 months
- Oral medication that increases SMN protein production

Multidisciplinary Clinics

- Essential for comprehensive care, including physical medicine and rehabilitation, neurology, pulmonology, orthopedics, nutrition, and physical therapy

Respiratory Care

- Regular monitoring of respiratory function
- NIV for patients with respiratory insufficiency
- Airway clearance techniques and cough-assist devices to manage secretions

Orthopedic Care

- Regular assessments to monitor and manage scoliosis and joint contractures
- Surgical interventions may be necessary for severe scoliosis

Genetic Counseling

- Should be offered to families to discuss inheritance patterns, family planning, and genetic testing options

Palliative Care

- Engage palliative care early to address quality of life and symptom management
- Focus on holistic care, including emotional and psychological support for patients and families

after onset, whereas the *A4V* mutation is typically associated with rapid decline and death within 1 year of onset.^{22,23} Since the identification of *SOD1* mutations over 20 years ago, numerous other ALS-related genes have been identified. Mutations in *C9ORF72* are the most frequently identified causative mutation in fALS and sALS, and they are responsible for approximately 40% of fALS and approximately 5% of sALS.^{25–27} There are currently over 30 gene mutations associated with the clinical syndrome of fALS, and because of rapid advances in genetic technology, the molecular understanding of sALS and fALS is constantly expanding.

As more genes responsible for fALS and sALS are identified, our understanding of the disease's underlying pathophysiology will improve. This advancement will enhance our ability to diagnose ALS, provide accurate prognoses, and design future therapies for patients. Clinically, fALS is similar to sALS but may be characterized by a younger average age of onset.²⁹ It may progress faster or slower than sALS depending on the pathogenic mutation.³⁰ Furthermore, targeting the genetic cause of ALS offers treatment strategies.

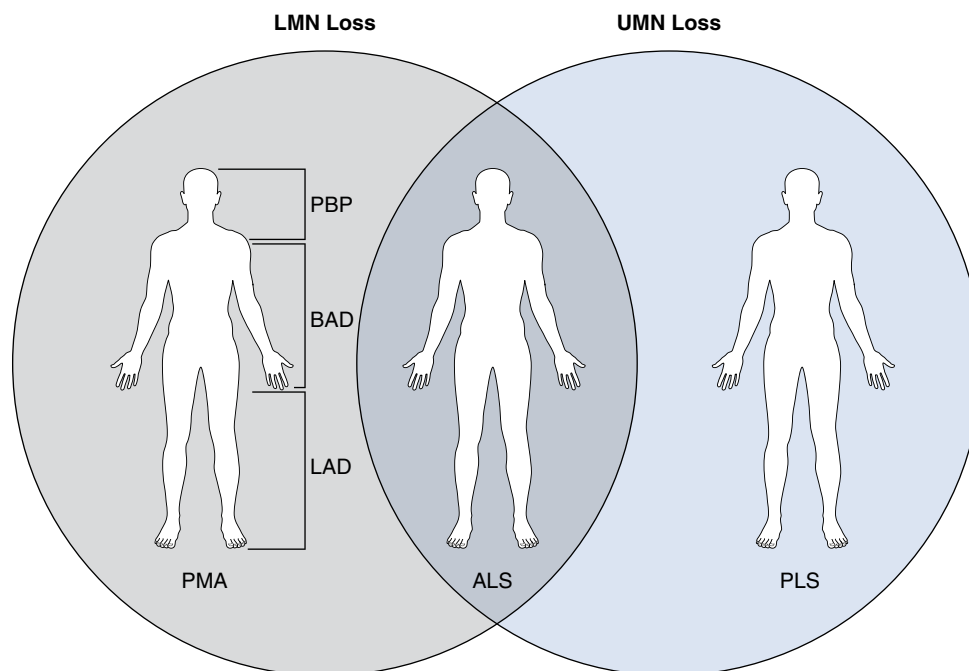
ALS Variants

ALS involves both the UMNs and LMNs, but some variant forms of ALS may be restricted to only UMNs or LMNs or certain body regions. Such variants include primary lateral sclerosis (PLS) and progressive muscular atrophy (PMA), where neuron loss is restricted to the UMNs or LMNs, respectively (Fig. 41.1). Similarly, there are regional variants, including PBP affecting the cranial nerve nuclei of the bulbar musculature, brachial amyotrophic diplegia (BAD) affecting the upper limbs in a proximal predominant

manner, and leg amyotrophic diplegia (LAD) predominantly affecting the distal lower limbs.^{56,103} Restriction to certain motor neuron pools or regions is often associated with a different prognosis. Importantly, in many if not most cases, there is subsequent progression to generalized involvement of both UMNs and LMNs typical of classic ALS. The relationship between ALS and other variants is not entirely clear, but there is strong evidence that these disorders are, at a minimum, tightly related. Phenotypic variability in genetically defined cases of fALS related to single mutations suggests that variants of ALS may be merely different faces of the same underlying pathology.

Although the pathologic relationship between ALS and ALS variants remains uncertain, distinguishing between these variants is helpful for clinicians. All restricted variants, except for PBP, have a better prognosis than typical ALS. Patients with PLS, PMA, and PBP may not be candidates for clinical trials for emerging ALS treatments. Thus, despite their similarities, it is useful for clinicians and patients to cautiously use these diagnostic categories when applicable. Generally, the clinical evaluation and treatment of patients with ALS and its variants are similar.

Primary lateral sclerosis. PLS is a disorder associated with progressive spasticity and weakness of limb and bulbar muscles related to degeneration of UMNs. PLS is rare and typically seen in the fifth decade. Similar to other sporadic MNDs, the etiology is not known. PLS is defined as having little or no LMN involvement clinically or by EMG. In spite of this, most patients with PLS eventually have fasciculations and cramps, some evidence of LMN loss on examination or EMG, and pathologic evidence of LMN involvement.¹¹² Although there are clear similarities with ALS, PLS progresses much more slowly than ALS—studies show an average lifespan of between 8 and 15 years after diagnosis.^{160,175}



• **Fig. 41.1** Spectrum of amyotrophic lateral sclerosis (ALS) and ALS variants. Progressive muscular atrophy (PMA) is associated with lower motor neuron (LMN) features of hypotonic muscle weakness, atrophy, fasciculations, flaccid dysarthria, and absent or reduced reflexes. Primary lateral sclerosis (PLS) is associated with weakness with incoordination or slowness of movements, brisk reflexes, spasticity, pathologic reflexes, and spastic dysarthria. ALS has features of both upper motor neuron (UMN) and LMN loss. PMA can be further divided into disorders that are restricted to certain regions, including progressive bulbar palsy (PBP) affecting the bulbar muscles, brachial amyotrophic diplegia (BAD) affecting the upper limbs, and leg amyotrophic diplegia (LAD) affecting the lower limbs. (Courtesy Chad Myers).

Although it may not be entirely clear that ALS and PLS are different diseases, given the distinct clinical features and prognosis, it is as Charcot wrote: “the clinical description deserves to exist alone.”³⁷

Most patients with PLS initially present with unilateral leg spasticity that involves the other leg within approximately 1 to 2 years. Signs and symptoms progress to the upper limbs within the next 3 to 4 years, and eventually there is pseudobulbar involvement over the next 1 to 2 years.²⁰⁰ Mills syndrome, a hemiplegic variant of PLS originally described in 1900, is associated with a unilateral presentation of UMN signs in the upper and lower limbs.¹³² The primary findings on examination and the cause of disability in PLS are spasticity and clumsiness rather than overt weakness, consistent with an UMN syndrome.^{112,200} Eventually, bulbar dysfunction is seen in nearly all patients. Unlike ALS, symptoms of bladder dysfunction are common late in PLS.^{112,200}

The diagnosis of PLS is established clinically, but testing can be helpful to exclude other diagnoses. Evaluation with magnetic resonance imaging (MRI) should be performed primarily to exclude other central nervous system disorders, such as multiple sclerosis; however, cortical atrophy and increased T2 signal intensity of the pyramidal tracts are often seen in PLS.^{112,160} Severe abnormalities or absent potentials on transcranial magnetic stimulation (TMS) are seen in nearly all cases of PLS and may confirm the diagnosis, but TMS is not widely available.^{109,112,160} By definition, electrodiagnostic studies should not show significant LMN loss, but minor changes can be seen. The differential diagnosis for PLS includes other central nervous system disorders; ALS and hereditary spastic paraparesis are two such important conditions. The use of genetic analysis should be considered for hereditary spastic paraparesis, particularly in patients with relatively symmetric lower limb involvement. Electrodiagnostic studies should be performed to exclude LMN involvement, which if found may support or indicate ALS.

Progressive muscular atrophy. PMA is a MND marked by LMN degeneration, causing progressive weakness and muscular atrophy. PMA is rare and affects primarily older males. By definition, patients with PMA have no clinical evidence of UMN dysfunction at onset, but it is important to note that as many as 70% of patients with PMA will eventually demonstrate signs of UMN degeneration.¹⁸⁴ Techniques such as TMS and MRI spectroscopy can show subtle corticospinal tract involvement in over 50% of patients with PMA. In one autopsy series, nearly every patient showed ubiquitinated inclusions in LMNs, which is a pathognomonic sign of ALS.⁹³ It remains controversial whether PMA represents a disease distinct from ALS. Nevertheless, the general approach to PMA is similar to ALS, but because of the different presentation, treatment, and prognosis, it should be considered as a related but possibly distinct disorder. Patients with PMA show atrophy and weakness, beginning in the hands in nearly 50% of cases, and less commonly in the lower limbs, shoulder girdle, and bulbar musculature.³⁹ Fasciculations are usually evident on examination. Reflexes are reduced or absent, and features of UMN dysfunction are absent.

The approach to diagnosis and treatment of PMA is similar to that of ALS (both described in detail later). EMG abnormalities consist of fibrillation potentials, positive sharp waves, fasciculation potentials, large MUAPs, and reduced MUAP recruitment. The differential diagnosis includes LMN-predominant ALS, other MNDs, and MMN. MMN is particularly important to exclude because this disorder is treatable with immunomodulation, such as immunoglobulin therapy. Prognosis is better than ALS, with a

5-year survival rate of over 50%, and patients live an average of 1 year longer than patients with ALS.^{39,107} Older age, lower forced vital capacity (FVC), lower ALS Functional Rating Scale (ALSFRS), and widespread involvement are poor prognosticators.^{39,107}

Regional ALS variants. The typical course of ALS involves focal onset and subsequent regional spread over time. Importantly, it is common for patients to have unrecognized mild or even moderate motor deficits in other regions, and at presentation the disease may be much more widespread on examination than recognized by the patient. During the natural course of ALS, most patients will develop generalized disease, but certain variants of ALS may remain restricted to certain regions or populations of motor neurons. Some of these variant syndromes include PBP, BAD, and LAD. PBP is an ALS variant with exclusively bulbar symptoms.¹⁴⁴ As mentioned, some patients may develop significant limb involvement and eventually may be characterized as bulbar-onset ALS, but this is a less common transition (7.5%) than in PLS or PMA.⁸⁸ PBP comprises almost 10% of all MNDs, primarily affecting females. PBP has a significantly older average age of onset and carries a poorer prognosis compared with typical ALS.³³ BAD is a LMN syndrome that appears to be closely related to ALS.¹⁰³ Patients with BAD typically have prominent proximal upper limb weakness that begins unilaterally and spreads to the contralateral limb. Distal upper limb muscles are relatively spared, and usually finger extensor muscles are more severely affected than finger flexors. Muscles of the bulbar, thoracic, and lumbar regions may remain unaffected for a decade or more, although in some cases there is more rapid progression. LAD is a lower limb correlate of BAD, but weakness tends to start in an asymmetric, distal-predominant manner with subsequent spread to the contralateral lower limb. Similar to BAD, LAD may have delayed progression to other spinal regions and therefore a better prognosis. Interestingly, patients with known causative ALS-related mutations may present with these variant syndromes, further supporting a close relationship between ALS and ALS variants.¹⁸⁷

ALS-plus syndromes. Although ALS is usually associated with isolated motor deficits, involvement outside of the motor system can occur. By definition, ALS-plus syndromes meet clinical and electrodiagnostic criteria for ALS, but they also have associated nonmotor neuron features that may include parkinsonism, FTD, ocular motility abnormalities, extrapyramidal signs, autonomic dysfunction, or sensory loss.^{15,169} Certain forms of fALS have been associated with the development of nonmotor features. For instance, patients with the TAR DNA-binding protein (*TAR-DBP*) gene mutations, known to cause fALS, have been reported with clinical features of parkinsonism.⁶⁹ Both FTD and fALS are seen in isolation and combination in patients and families with *C9ORF72* gene mutations.^{51,163} The association of nonmotor features in some patients with sporadic and fALS strongly suggests that ALS is a multisystem disease with predilection for UMN and LMN involvement.

ALS Assessment

Clinical presentation. Clinical presentations of ALS can vary greatly, but almost always include painless weakness and atrophy. The initial location of the weakness divides roughly into thirds: one-third in the bulbar region, one-third in the upper limb, and one-third in the lower limb.³³ In the limbs, weakness usually starts in an asymmetric and distally predominant fashion.⁸⁸ The weakness seen in ALS often begins focally and then generalizes over time, though subtle weakness may be present in other areas the patient is not aware of. For instance, symptoms of foot drop or

hand weakness may prompt a clinical visit, and then more generalized weakness, fasciculations, or hyperreflexia are noticed on exam. Patients may not clearly articulate complaints of weakness, but rather may describe difficulties with certain activities, such as climbing stairs, walking, or opening a jar. Sensation is not affected, and the presentation of weakness with no numbness and tingling is a relatively rare clinical scenario that should raise red flags that MND or another pure motor process is present.

Failure of motor neurons leads to many clinical features outside of the limbs. Bulbar involvement in ALS at presentation is twice as common in females than in males and more common in older patients.^{40,88} Although around one-third of patients have bulbar symptoms as a chief complaint, the majority have such symptoms at presentation when specifically asked.⁸⁸ Those presenting with bulbar symptoms nearly always have dysarthria.⁸⁸ Dysphagia is a rare presenting complaint but is expected during the course of disease. Nearly one-quarter of patients acknowledge respiratory symptoms when specifically addressed.⁸⁸ These may include sleep apnea, exertional dyspnea, or orthopnea. Isolated respiratory insufficiency at presentation is exceedingly rare; when respiratory involvement is noted at presentation, there are usually other generalized deficits that are also apparent on examination.^{33,40} Presenting with isolated complaints of fasciculations is similarly rare, but it is not rare for fasciculations to be noted during the initial exam.⁸³ Pain and other sensory symptoms are rare at the time of diagnosis and suggest an alternative diagnosis. An exception is cramping, which is the presenting complaint in around 10% of patients.

Spasticity, clumsiness, increased muscle stretch reflexes, and the Hoffmann sign are evidence of UMN degeneration. The Hoffmann sign in the upper limb and Babinski sign in the lower limb are useful signs of UMN pathology. In one series, 94% of patients with ALS and a positive Hoffmann sign had corticospinal tract degeneration on autopsy.¹⁹⁵ The Babinski sign is less reliable, with only one-quarter of patients with ALS having a positive response bilaterally on presentation.⁸³ Importantly, a normal reflex in an atrophic limb may represent an exaggerated reflex response. An exaggerated jaw jerk, pseudobulbar affect, positive snout, or palmomental reflexes can help establish UMN dysfunction in bulbar musculature. In ALS, there is typically a combination of UMN and LMN signs, starting asymmetrically and progressing to involve multiple regions. Increased reflexes are seen on presentation in approximately one-half of patients with ALS.⁸³

Laboratory tests for ALS. Laboratory testing is important in the diagnosis of ALS to assess for potential mimics, but no test currently available is specific to diagnose ALS. Importantly, some laboratory testing may show nonspecific abnormalities in patients with ALS, which can cause confusion. For instance, CK is often elevated, but typically less than 10 times the upper limit of normal.²⁸ CSF protein may also be elevated, but rarely over 60 mg/dL.⁸⁸ Anti-GM1 antibodies are seen in MMN, an important disorder in the differential of focal/multifocal weakness without sensory loss or pain. The presence of these antibodies can help support the diagnosis of MMN but has been shown to be positive in up to 10% of patients with ALS.⁸⁸ This can pose a problem in patients without UMN signs, in whom MMN is an important consideration. In patients suspected of having ALS, genetic testing can be helpful to determine the presence of a pathogenic mutation attributable to fALS. Outside of genetic testing in fALS, there are no laboratory tests that can establish the diagnosis of ALS. Similarly, in a patient whose clinical and electrophysiologic profile is strongly suggestive of ALS, there is no laboratory study that can effectively rule it out.²⁸

Electrodiagnostics in ALS. Electrodiagnosis is the main test used to diagnose ALS and should be done in nearly all cases where ALS is suspected. Abnormalities on needle EMG have the same significance as clinical evidence of LMN disease in the Gold Coast diagnostic criteria (see later). A combination of signs of active denervation (fibrillation potentials, positive sharp waves, and fasciculations) and chronic denervation (high amplitude and wide duration MUAPs) are seen in most patients. Unstable MUAPs on triggered analysis are evidence of chronic denervation and immature synapse formation. Reduced recruitment is expected. The distribution of these findings correlates with the clinical deficits and usually progresses over time. Therefore, targeting clinically involved muscles increases the yield of needle EMG. Needle EMG is particularly useful to show subtle evidence of LMN disease in areas where it is not clinically evident. Two muscles of different root and peripheral nerve supply must be involved to demonstrate LMN degeneration in a given region.²⁷ Early in the course of disease, there is focal LMN degeneration, and the majority of muscles on clinical and EMG assessments may be normal, making the diagnosis of ALS more challenging or impossible. Thus, a high index of suspicion is needed to identify the disease earlier in the course. Motor NCS will show low amplitudes where significant axon loss has occurred but should otherwise be relatively normal. Evidence of conduction block suggests alternate, more treatable diagnoses such as MMN. Sensory conduction studies are usually normal, but coexistent sensory abnormalities can occur in ALS in up to 12% of patients, in which case findings are usually very mild compared with motor involvement.⁸⁸

Imaging in ALS. Imaging is frequently obtained in patients with suspected ALS. Corticospinal tract abnormalities can be seen on brain MRI but are nonspecific and not usually helpful diagnostically. MRI of the cervical spine can help to rule out structural lesions associated with cord compression or syringomyelia. It is worth noting that incidental findings of mild to moderate cervical spondylosis on imaging are not infrequent, and features of ALS should not be mistakenly attributed to clinically insignificant focal structural lesions. Furthermore, lack of sensory symptoms to corroborate myelopathy or radiculopathy may favor ALS. MRI of the brain is also commonly performed, particularly in a patient with bulbar or pseudobulbar symptoms or signs. Similar to cervical spine and incidental degenerative changes, approximately 10% of patients ultimately diagnosed with ALS will have incidental findings of nonspecific ischemic changes on a brain MRI.⁸⁸ In circumstances of focal cervical spine disease and mild ischemic changes identified on MRI, features of LMN loss (outside of the cervical region in the case of cervical spine disease) support the diagnosis of ALS as opposed to another process affecting the central nervous system.

ALS diagnostic criteria. Diagnostic criteria have been developed to assist clinicians and researchers in identifying patients with ALS. To overcome the limitations of previous categories, a consensus group of international ALS experts developed the Gold Coast criteria, which are simpler and more reflective of clinical practice (Box 41.3). The key components of the Gold Coast criteria comprise (1) progressive motor impairment confirmed by either history or repeated clinical assessments, after an initial period of normal motor function; (2) the presence of both UMN and LMN dysfunction in at least one body region (with both UMN and LMN dysfunction in the same region if only one region is affected) or LMN dysfunction in at least two body regions; and (3) the exclusion of diseases that mimic ALS through appropriate investigations.¹⁷¹

• BOX 41.3 Gold Coast Criteria for the Diagnosis of Amyotrophic Lateral Sclerosis

1. Progressive motor impairment documented by history or repeated clinical assessment, preceded by normal motor function, and
2. Presence of upper^a and lower^b motor neuron dysfunction in at least 1 body region^c (with upper and lower motor neuron dysfunction noted in the same body region if only one body region is involved) or lower motor neuron dysfunction in at least 2 body regions, and
3. Investigations^d excluding other disease processes.

^aUpper motor neuron dysfunction implies at least one of the following:

1. Increased deep tendon reflexes, including the presence of a reflex in a clinically weak and wasted muscle, or spread to adjacent muscles
2. Presence of pathologic reflexes, including Hoffmann sign, Babinski sign, crossed adductor reflex, or snout reflex
3. Increase in velocity-dependent tone (spasticity)
4. Slowed, poorly coordinated voluntary movement, not attributable to weakness of lower motor neuron origin or parkinsonian features

^bLower motor neuron dysfunction in a given muscle requires either:

Clinical examination evidence of muscle weakness, and muscle wasting or

Electromyography (EMG) abnormalities that must include:

Both evidence of chronic neurogenic change, defined by large motor unit potentials of increased duration and/or increased amplitude, with polyphasia and motor unit instability regarded as supportive but not obligatory evidence.

And evidence of ongoing denervation, including:

Fibrillation potentials or positive sharp waves, or fasciculation potentials

^cBody regions are defined as bulbar, cervical, thoracic, and lumbosacral. To be classified as an involved region with respect to lower motor neuron involvement, there must be abnormalities in two limb muscles innervated by different roots and nerves, or one bulbar muscle, or one thoracic muscle either by clinical examination or by EMG.

^dThe appropriate investigations depend on the clinical presentation, and may include nerve conduction studies and needle EMG, magnetic resonance imaging or other imaging, fluid studies of blood or cerebrospinal fluid, or other modalities as clinically necessary.

In a recent cohort study, the Gold Coast criteria showed significantly higher sensitivity than both the prior Awaji and revised El Escorial definite/probable ALS criteria, underscoring their practical value.¹⁷² Although the Gold Coast criteria showed a high level of sensitivity, highlighting the effectiveness of clinical assessment in identifying UMN dysfunction, there is room for improvement in ALS diagnosis through objective assessment of UMN dysfunction, an aspect that merits exploration in future prospective studies. The essence of the criteria is that discovering both UMN and LMN findings in the same area is rare and suspicious for ALS. The more areas where these findings are identified, the more likely the diagnosis is ALS, and the less likely these findings can be realistically explained by any other disorder.

General approach to diagnosis of ALS. The overall approach to a patient with suspected ALS is complex and begins by establishing which type of motor neurons and which body regions are involved. A pragmatic approach is to first assess if there is clinical bulbar or pseudobulbar involvement. If so, an MRI of the brain should be obtained. If cervical, thoracic, and lumbar regions are involved, spinal MRI should be pursued or at least considered. This can help exclude the most common alternate diagnoses that are usually evident on neuroimaging, but the interpretation of minor, common MRI abnormalities needs to be done with caution. The differential can then be narrowed by the type of motor neurons that are clinically involved. Almost all patients should have an EMG performed. This helps determine the pattern of LMN involvement more accurately and makes the diagnostician more confident about the group of diagnoses to pursue.

The possible causes of both UMN and LMN injury in the same patient is a short list of rare conditions. A relatively common exception is cervical radiculomyelopathy related to degenerative cervical spine disease. A patient with this condition will have LMN injury and findings at the level of the radiculopathy, and UMN signs caudal to this level attributable to the myelopathy. A representative patient could have C6 cord compression from spondylosis. The resultant radiculopathy could lead to weakness and atrophy in the C6 myotome and diminished biceps and brachioradialis reflexes. The myelopathy would cause an increase in the triceps and lower limb reflexes, Hoffmann and Babinski signs, and lower limb weakness or clumsiness without atrophy. This is a reasonably common and treatable finding in clinical practice, making it important to exclude. A syrinx could cause a similar syndrome but is much less common. Another common mimicker is a cervical myelopathy associated with cervical spine degeneration and superimposed length-dependent neuropathy. The myelopathy would cause UMN findings in the upper limbs, but in the lower limbs the neuropathy often blunts reflexes to make them normal or reduced. In this case, the LMN findings are distal to the UMN findings, and the rule for rostral findings of UMN signs discussed earlier is not helpful. In both situations, the sensory complaints and findings are often clearly evident. A useful general rule is that surgically correctable myelopathies or radiculopathies related to cervical spondylosis usually have more prominent sensory signs or symptoms. These clinical possibilities are both easily excluded with an MRI of the cervical spine. Other rare possible causes of a combined UMN/LMN presentation include spinal dural arteriovenous malformations, hereditary motor neuropathy, hereditary spastic paraparesis, and other inherited disorders.

A patient with signs, symptoms, and testing consistent with LMN injury without clear UMN involvement has a greater number of possible causes that need to be excluded. For instance, myasthenia gravis, other NMJ disorders, and MMN are important because these are treatable diseases. PMA, adult-onset SMA, inclusion body myositis, HD, and KD are other possibilities. Electrodiagnostic testing is a critical part of the evaluation, which needs to be designed to address potential mimickers. For instance, an electrodiagnostic study may need to include multiple motor NCS with multiple stimulation sites to investigate possible conduction block, as evidence for MMN. Repetitive stimulation and occasionally SFEMG may be considerations. Clinicians need to be wary of false positives as a source of diagnostic confusion in patients with ALS.

A patient with primarily or exclusively UMN signs has quite a different list of possible diagnoses. These include inflammatory conditions, such as multiple sclerosis and neuromyelitis optica. These should be excluded by the rate of progression, sensory complaints, and MRI of the brain and spinal cord. Other myelopathies must be considered, including compressive, malignant, vitamin B12 deficiency, copper deficiency, and syringomyelia—all of which can be easily excluded with imaging and laboratory testing. When imaging and laboratory testing are negative, PLS is high in the differential, but hereditary causes, such as hereditary spastic paraparesis, Silver syndrome (hereditary motor neuropathy V), or a female carrier of X-linked adrenoleukodystrophy/adrenomyeloneuropathy, can remain as possibilities.

Patients with predominantly bulbar symptoms have a shorter list of alternate diagnoses. Myasthenia gravis, KD, and myopathies such as oculopharyngeal muscular dystrophy are primary considerations. EMG testing is critical and needs to include

• BOX 41.4 Key Points in the Diagnosis of Amyotrophic Lateral Sclerosis

History

- Focal weakness without pain or sensory complaints is suspicious for amyotrophic lateral sclerosis (ALS)
- Patients with sensory complaints or isolated fasciculations rarely have or develop ALS

Physical Examination

- Upper motor neuron (UMN) and lower motor neuron (LMN) findings in the same limb are very suspicious for ALS
- Fasciculations in isolation are common in healthy patients and in many LMN processes
- In combination with an abnormal examination or electromyography (EMG), fasciculations are highly suspicious for ALS or other motor neuron diseases (MNDs)
- When seen with otherwise normal examination findings and EMG, fasciculations are almost always benign

Testing

- Electrodiagnostic testing is the cornerstone of identification of LMN loss in all MNDs, but diagnosis of ALS requires clinical correlation with examination features of UMN involvement
- No laboratory or imaging test can make the diagnosis of ALS in isolation, but genetic testing can identify pathogenic gene mutations associated with familial ALS and in some cases of sporadic ALS
- In a patient with clinical features of ALS, no diagnostic test can rule out ALS
- Most patients suspected of having ALS should undergo testing to exclude mimicking conditions
- Some testing results can be nonspecific. For example, patients with ALS often have mildly elevated creatine kinase and occasionally elevated anti GM-1
- In most patients, imaging of the brain and cervical spine are particularly important to exclude other causes of UMN or LMN dysfunction

repetitive stimulation of facial muscles and possibly SFEMG. Findings of NMJ abnormalities on electrodiagnostic testing should be interpreted in the context of the presence or absence of denervation because secondary decrement on RNS or jitter and blocking on SFEMG are common in ALS.

Diagnosing ALS must be done with caution and usually confirmed by a clinician with expertise in MND (Box 41.4). There is a high price for both missing the diagnosis and giving it falsely. Giving the diagnosis of ALS incorrectly (a false-positive diagnosis) is estimated to occur in 5% to 10% of cases.^{48,183} About half of these cases had a treatable diagnosis, usually MMN or a cervical spondylotic myelopathy.⁴⁸ KD was missed in nearly 15% of false-positive ALS diagnoses. Facial fasciculations and a family history were both evident in 75% of cases and were paramount in making the correct diagnosis. Gynecomastia and tremor, findings expected in KD, were both only seen in about half of patients and were less reliable than might be expected.¹⁸³ Clinicians should be on the lookout for red flags that the diagnosis of ALS is incorrect, including development of atypical symptoms or a failure to progress as expected.^{48,183}

Missing the diagnosis of ALS (a false-negative diagnosis) also has consequences and is significantly more common. Nearly half of patients who are ultimately diagnosed with ALS were initially misdiagnosed.²¹ The most common incorrect diagnoses were lumbar and cervical stenosis. Spondylosis is very common in this age group and is seen to some extent in most patients who are

ultimately felt to have ALS. The provider must therefore be cautious to avoid quickly assuming a causal relationship between common minor findings of spinal degeneration and often dramatic neurologic findings that are disproportionate and are poorly correlated. Unfortunately, around 10% of patients ultimately diagnosed with ALS had spinal surgery before the ultimate diagnosis of ALS—an example of the cost of this mistake.²⁰ Other common misdiagnoses include cerebrovascular disease in about 10% and “nothing wrong” in around 15%.²⁰ A total of ~30% of patients ultimately diagnosed with ALS had an unnecessary surgery for an incorrect diagnosis.²⁰ An estimated cost for misdiagnosis is \$30,000.²⁰ A loss of faith in the medical system and the emotional toll for patients and families are harder to quantify.

It is therefore difficult to decide when exactly the suspected diagnosis of ALS should be given. A level of certainty needs to be established, and a particularly high degree of certainty that a treatable diagnosis is not being missed is critical. Waiting until there is absolutely no question of the diagnosis can have negative consequences as well. Giving the diagnosis early has psychological, economic, and medical benefit.⁸ The manner in which the diagnosis is given is important as well and has lasting consequences to the patient. Experts recommend that a doctor who knows the patient give the diagnosis in person with distractions, such as the cell phone and incoming interruptions from staff, eliminated and with 45 to 60 minutes available. It is helpful to begin with what the patient knows to determine the best way to communicate with the patient. If family is available for the clinic visit and the patient is agreeable to having family be part of the discussion, this is usually very beneficial. The diagnosis and what it will mean to the patient should be rolled out in a stepwise manner, making sure each step is understood before moving on. Providing additional written information or a recommended website is important. It is most important that the patient be assured in words and in actions that support is at hand, and setting up a follow-up visit in 2 to 4 weeks before the patient leaves the office reinforces the provider's commitment to the patient. Common mistakes include withholding the diagnosis, delivering it without empathy, and not giving enough information. The patient needs to be left with hope, and phrases like “there is nothing we can do” are inaccurate and unproductive. Giving the diagnosis with improper technique can lead to a sense of abandonment, erode the patient-physician relationship, and affect the patient's psychological adjustment to bereavement.⁸

Genetic counseling and testing should be considered for patients with newly diagnosed ALS, even in those without a known family history. For patients with a first- or second-degree family member with ALS or FTD, the yield of finding a highly penetrant, causative mutation is 60% to 70%.¹⁶⁵ The clinical implementation and usefulness of genetic testing is rapidly evolving. Specifically, genetic therapy for *SOD1*-related fALS is now clinically available. This makes genetic testing crucial for identifying potentially treatable forms of ALS, as new treatments targeting specific mutations are developed and tested in clinical trials. In some cases, if a specific genetic mutation is identified, this may help provide prognosis and information about the potential rate of disease progression. The information gained through genetic testing may be useful for unaffected relatives. Nevertheless, respecting the right of an asymptomatic relative's desire not to know the possible genetic risk needs to be respected. Some also advocate genetic testing for those without a known affected family member. The likelihood of finding a known causative mutation in sALS is 11%, but for those with ALS and FTD it approaches

90%.¹⁶⁵ The *C9ORF72* mutation is responsible for the majority of identified genetic causes of sALS, both with and without FTD.¹⁶⁵ Surveys of patients diagnosed with ALS who have undergone genetic testing show the vast majority of respondents (>80%) would recommend testing to others.¹⁸⁹

ALS Functional Assessment

The revised version of the ALSFRS (ALSFRS-R) is the most commonly used tool to assess functional deficits in ALS longitudinally.^{14,32} This disease-specific scale consists of 12 items grouped into four domains that evaluate bulbar symptoms, limb and trunk functionality, respiratory symptoms, and the necessity for interventions such as percutaneous endoscopic gastrostomy (PEG), noninvasive ventilation (NIV), or tracheostomy with invasive ventilation.³² Although assessments were originally conducted exclusively through in-person and telephone interviews with health care professionals, self-assessments and online questionnaires are now widely used.^{101,104,135} The ALSFRS-R, initially developed for clinical trials, has become a standard tool in ALS research and practice because of its strong correlation with disease progression and survival. However, a significant challenge is that individuals with identical total scores may have different disease stages or prognoses. This issue, known as multidimensionality, arises from combining diverse subscales such as bulbar, motor, and respiratory functioning.⁶⁷

ALS Treatment

Disease-modifying medications. Three medications are currently available for disease modification in the treatment of ALS: riluzole, edaravone, and tofersen. Riluzole and edaravone have been shown to slow the progression of sALS and are approved for clinical use. Tofersen is specifically available for fALS related to *SOD1* mutations. Generally, the treatment of ALS variants involves using disease-modifying agents approved for sALS.

Riluzole. Multiple studies have shown riluzole to be safe and effective in slowing the progression of the disease.¹²⁸ The precise mechanism of effect of riluzole is unclear. Riluzole reduces mortality 23% at 6 months and 15% at 12 months and appears to prolong survival by around 4 months.¹⁸² There is a clear dose effect.⁸ The best benefit-to-risk ratio is achieved at the standard dose of 50 mg twice daily. There are few serious side effects. The most common adverse reactions are dizziness, asthenia, and gastrointestinal disorders. Liver function test abnormalities are common and mild, with significant elevations of over five times the normal limit seen in less than 5% of patients.¹¹¹ Routine monitoring of liver function tests is recommended. Our typical protocol includes testing at baseline, monthly for 3 months, and then every 3 months for 1 year. Riluzole should be given as early as possible because the life-prolonging effect seems to be greater when started earlier.⁸ Expense is the most common barrier to its use; other barriers include a perceived lack of efficacy, lack of information about the drug, concern about side effects, and in some cases the treating physician recommending against it.²⁶ It is paramount that proper information is provided so that patients can make well-informed decisions.

Edaravone. This free radical scavenger was originally developed for the treatment of stroke that was shown to have efficacy in a double-blind, placebo-controlled trial of over 200 patients with ALS.¹ There was no statistically significant difference in the primary endpoints of the trial, but a subgroup of the patients treated seemed to benefit, leading to a second smaller trial of patients with characteristics similar to those of the subgroup that benefitted in

the first trial. In this select group of 130 patients, a significant slowing of the disease by approximately 30% was seen over a 6-month period, which has generated great enthusiasm. In spite of a sizable effect and favorable side effect profile, the interest in the patient community has been met with caution in the medical community because the results are based on one small trial of a select group of patients with no benefit in a broader spectrum of patients with ALS.¹ This group that benefitted represents a small proportion of ALS patients; these patients all had mild to moderate dysphagia, fine and gross motor dysfunction, were treated within 2 years of symptom onset, and had normal respiratory function. However, survival data from randomized clinical trials are lacking, and retrospective findings from other studies are inconsistent.¹⁹³ The originally approved form of edaravone is delivered intravenously, but the oral formulation medication has been preferred over injections or infusions due to fewer office visits and the avoidance of injection-site reactions and other side effects.^{11,71} The recommended oral dosage is 105 mg (5 mL) every morning for 14 days, followed by a 14-day drug-free period. For subsequent treatment cycles, the same dosage is administered on 10 of 14 days, followed by another 14-day drug-free period. Oral edaravone should be taken in the morning on an empty stomach after overnight fasting, with no food or drink (except water) for 1 hour. Patients should fast for 8 hours before each dose if they have consumed a high-fat meal, 4 hours if they have consumed a low-fat meal, or 2 hours if they have consumed a caloric supplement. Additionally, oral edaravone can be administered orally or through a nasogastric (NG) or PEG tube.^{73,153}

Tofersen. Genetic-based therapies are an emerging area in ALS therapeutics. As genetic causes of fALS are determined, this opens possibilities for novel therapies targeting genetic mechanisms of MNDs. The first genetic-based therapy for fALS, tofersen (BIIB067) for treating *SOD1*-ALS, was granted accelerated approval in 2023.¹⁰ Tofersen is an antisense oligonucleotide designed to target and reduce the production of the *SOD1* protein, which is associated with ALS development.¹³⁰ The recommended intrathecal dose of tofersen is 100 mg/15 mL per administration, with initiation of treatment consisting of three loading doses administered at 14-day intervals, followed by a maintenance dose every 28 days thereafter. In the phase 3 trial (VALOR) involving 108 ALS patients, tofersen did not demonstrate a significant difference in the decline on ALSFRS-R compared with placebo. However, secondary endpoints indicated a reduction in CSF *SOD1* protein and plasma neurofilament levels in the faster-progressing subgroup. The open-label extension study suggested that earlier initiation of tofersen might lead to better outcomes. Common adverse events included lumbar puncture-related adverse events, headache, pain in the arms or legs, falls, and back pain.¹³¹ Further evaluation is needed to fully understand the efficacy and long-term effects of tofersen in ALS treatment. As a result, a phase 3 study (ATLAS) is currently underway.²²

Rehabilitation, symptomatic and medical treatment. Strategies for rehabilitation must be designed for the targeted patient population based on disease impact and natural history. In particular, the progression of ALS makes prescribing rehabilitation interventions for patients difficult because of a changing functional status. Additionally, the combination of spasticity and severe focal weakness creates a challenge for those trying to maximize function and ease care in this population.

Stages of Rehabilitation. Patients with ALS and primarily spinal involvement tend to progress over six stages with common clusters of rehabilitation needs in each, though individuals can

vary considerably and straddle stages rather than fall neatly into one.¹⁷⁴ Outlining these stages and their challenges helps the provider anticipate the rehabilitation treatments they will likely be discussing at the next visit.

Stage 1 includes patients who are ambulatory and fully independent, with mild weakness or clumsiness. They need only to maintain active range of motion (ROM) and carry on with normal activities of daily living, but they can benefit from strengthening unaffected muscles. It is reasonable to strengthen all muscles, including affected muscles as long as the provider takes caution to prevent overwork damage (see later).

Patients in stage 2 remain ambulatory and independent but have moderate weakness. Functional impairment may be severe in some areas despite preserved overall function (Fig. 41.2). They often benefit from ankle-foot orthoses to compensate for foot drop. However, if the patient has plantar flexion spasticity, a dorsiflexion assist can encourage clonus and should be avoided. Anterior trim lines add weight but can provide medial-lateral



• **Fig. 41.2** Profound intrinsic hand muscle weakness in a patient diagnosed with amyotrophic lateral sclerosis with otherwise retained bulbar and ambulatory function. The patient is unable to voluntarily grip the pen to sign a consent form.

stability for those who require it. A wrist-hand orthosis to assist with wrist drop is often helpful. An opponens splint can be used to accommodate lack of thumb opposition because of thenar muscle weakness (Fig. 41.3). Hand weakness also commonly makes buttoning difficult, and assistive devices to ease this task are helpful. Focal weakness of the cervical extensor muscles leading to head drop or chin-on-chest deformity is a common problem in ALS. Cervical orthoses to stabilize the cervical spine may help provide some relief but are sometimes poorly tolerated. Work simplification and input from an occupational therapist can be useful at this stage. Exercise prescription is similar to stage 1.

Stage 3 includes patients who remain ambulatory but have severe weakness in selected muscle groups. A common problem that arises at this stage is the inability to rise from a chair. Lift chairs and elevated toilet seats can mitigate this problem. Ambulation aids such as a walker can be helpful, and a manual wheelchair for long trips is often necessary and facilitates socialization. This should be custom fitted to provide pressure relief and support the spine, but financial and insurance constraints must be considered when determining the most appropriate type of durable medical equipment. At this stage, deep breathing exercises are helpful for those who develop mild respiratory muscle weakness.

Stage 4 includes patients who are nonambulatory but remain independent. Electrically powered chairs are usually appropriate. Strengthening exercises are usually limited to any unaffected muscles, but ROM exercises for other areas are also important. A hospital bed facilitates bed mobility.

Stage 5 is entered when the patient is no longer independent. Transfer assistance with devices such as Hoyer lifts is generally required. Family education is a large part of the care of the patient. A wheeled shower chair is often helpful for these patients. A lack of mobility makes pressure relief measures in the wheelchair and bed higher priorities.

At stage 6, patients are completely immobile, dependent, and require maximal assistance from skilled caregivers. Proper positioning is important to keep patients as comfortable as possible and prevent pressure ulcers. Palliative care becomes the focus of treatment.

Spasticity. Spasticity can have a significant impact on the function of patients with predominantly UMN involvement. Treatment of spasticity in patients with ALS is similar to other diseases



• **Fig. 41.3** Orthotic intervention for severe intrinsic hand muscle weakness and impaired thumb opposition in a patient with amyotrophic lateral sclerosis.

of the central nervous system but with some important differences. Physical therapy has been shown to reduce spasticity in patients with ALS; oral baclofen is commonly used but has not been shown to have clear benefit.^{57,145} Intrathecal baclofen has been used successfully in ALS patients,¹²¹ but not all are appropriate candidates, specifically those with rapid progression or very advanced disease. Botulinum toxin can be useful in treating specific muscle groups as in other disorders, but it can lead to generalized weakness that can be more problematic in a patient with ALS than it is for someone with a traumatic brain injury or stroke.¹²⁷ Tizanidine, dantrolene, gabapentin, and benzodiazepines are commonly used but have not been specifically studied in ALS.¹⁷³ Benzodiazepines carry the additional benefit of helping treat anxiety, but they also carry the additional risk of respiratory depression. Dantrolene does not share this risk, but some advocate avoiding it in patients with ALS.

Communication. Patients with ALS can experience communication difficulties that may eventually lead to the inability to communicate independently. Dysarthria can result from both UMN and LMN dysfunction or a combination.¹⁷³ Other barriers patients with ALS face in effective communication include hypernasal speech caused by air leaking through the nose, hypophonia, and coexistent cognitive impairments that may occur in disorders such as combined ALS-FTD.

Effective communication is a major factor in the quality of life in patients with ALS and should be managed aggressively.^{8,138} Many patients elect to withdraw support when they become “locked-in” and unable to communicate, rather than when they lose a particular physical function. In addition to the inability to indicate needs, the main loss patients with impaired communication experience is of social closeness with their partner and family.^{138,173} Given this, patients should be routinely monitored by a speech-language pathologist to optimize communication and monitor changing needs during the course of the disease.⁸ Treatment options for impaired communication include palatal lifts or palatal augmentation prosthesis to decrease hypernasality.⁸ As communication becomes more limited, augmentative and alternative communications systems (AAC) are useful. These substantially improve the patient’s quality of life.⁸ AAC devices range from pointing boards to more complex eye-gaze devices or brain-computer interfaces. Some patients with ALS will utilize voice or message banking. In some cases, these services can be used to create a synthesized voice on AAC equipment that sounds similar to the patient’s own voice. Patients should be encouraged to consider this as soon as possible after their diagnosis so that they can use their strongest possible voice (<https://www.als.org/navigating-als/resources/fyi-voice-preservation>).

Dysphagia. Dysphagia is a common symptom in ALS. It contributes to malnutrition, poses a risk for aspiration, and can be a major source of anxiety. Dysphagia usually begins with thin liquids, and using thickener is a simple early solution. When dysphagia with solids begins, smaller, more frequent meals can be helpful. Moistening and softening solids can also make swallowing easier.¹⁷³ A speech therapist is a valuable team member and can teach and reinforce helpful strategies such as supraglottic swallowing, postural changes and using a chin tuck.⁸ Ultimately with disease progression, the failing swallowing mechanism needs to be bypassed if the patient is going to get safe and adequate nutrition.

Nutrition. Proper management of nutrition is one of the most important jobs of a physician managing a patient with ALS. Without intervention, eventually patients with ALS will not get enough calories, and this problem magnifies as the disease advances.¹⁰⁰ This is caused by a combination of factors: bulbar and

masticatory muscle weakness, dysphagia, fatigue, and slow eating. Patients with ALS also have increased resting energy expenditure.⁷ Weight loss is therefore common in patients with ALS. Almost three-quarters of ambulatory patients consume less than the recommended daily allowance of calories, and one-quarter have lost more than 10% of their body weight.¹⁷⁷ Malnutrition is proportional to the degree of dysphagia and is a major predictor of death in patients with ALS.^{53,177} Recruiting the help of a dietician can help patients maintain their calorie intake. Common interventions include modifications of food and liquid consistency, the chin-tuck maneuver, and high-caloric supplements. Carbonated liquids and ice-cold liquids are more easily consumed for some patients.⁷ Calorie intake and weight loss should be a part of every follow-up visit. The initial strategies mentioned earlier can relieve these symptoms in some patients with early ALS, but eventually to get proper nutrition and prolong survival, more definitive measures need to be taken.

An NG tube can provide greater nutrition to patients with dysphagia but is uncomfortable and can lead to aspiration if improperly placed. A PEG tube is a permanent solution, and there are excellent data to support the use of PEG tubes in patients with ALS. Use of a PEG tube helps stabilize body weight and probably prolongs survival, though it is not clear if it improves quality of life.^{102,122,129} Indications for placement of a PEG tube in a patient with ALS include decreased caloric intake with weight loss greater than 10% of baseline, dehydration, and meals limited by dysphagia or taking greater than 30 minutes.¹⁷³ Studies show improved outcomes if PEG placement is performed before the FVC drops below 50% of the predicted value.¹⁰² PEG tubes do not reduce the rate of aspiration pneumonia; as such, this is not an indication for placement.¹⁷³ Risks of PEG tube placement include tube dislodgement, benign pneumoperitoneum, parietal hematoma, peritonitis, gastrointestinal bleeding, gastrocolic fistula, and death.⁵² The overall risk of complications for PEG tube placement in patients with ALS is 6% to 10%.¹⁰² An alternative is a radiologically inserted gastrostomy (RIG) tube. The risks and survival benefits are the same as a PEG tube, but because it can be performed under local or light sedation (as opposed to deep sedation or general anesthesia), it can be done in patients with a lower FVC.¹⁷⁰ Home parenteral nutrition has been used for some patients with late-stage ALS.⁷

Despite the benefits of a PEG or RIG tube, most patients with severe dysphagia do not use them.²⁶ There are a variety of barriers that keep appropriate patients with ALS from using PEG or RIG tubes. Physicians recommend a PEG tube in less than half of the patients in whom it is indicated.²⁶ Less than half of the patients to whom it is recommended actually elect to have it placed.²⁶ When the physician brings up PEG tube placement to a patient with ALS, it is often received by the patient as a sign that things are going badly and death is near. This often leads to a delay in the discussion, at which point the risk of placing the PEG increases. Many patients ultimately decline a PEG tube because they are uncomfortable with the idea. Importantly, one study showed that over one-third of patients who did not receive a PEG tube lacked information about the role and potential benefits of a PEG tube or erroneously felt their swallowing was adequate.²⁶ Ultimately, many patients will refuse a PEG or RIG tube even with all the proper information, but the provider owes the patient the opportunity to make an informed decision.

Patients with ALS frequently look to alternative medicine for solutions that allopathic medicine is not providing. This is particularly true regarding nutrition. Nearly 80% of patients use

nutritional supplements or high-dose vitamins.²⁶ Unfortunately, none have shown benefit. Vitamin E and creatine have been studied most extensively and are not recommended to patients with ALS.¹²⁸

Sialorrhea. Sialorrhea in patients with ALS can be socially disabling and predispose them to oral infections.⁸ It is caused by impaired handling of saliva rather than overproduction; in fact, patients with ALS actually salivate less than healthy patients.³⁶ A home suction machine is something most patients find helpful. Multiple medications are commonly used to control saliva production in ALS. Amitriptyline is cheap, effective, and can be helpful in treating a concomitant mood disorder or pseudobulbar affect. Typical dose is 25 to 50 mg twice to three times daily.⁸ Atropine drops 0.5 or 1% sublingually three or four times daily can be used. Other treatments include nebulized or intravenous glycopyrrolate, oral or transdermal scopolamine, and benztropine.⁸ Side effects include constipation, urinary retention, confusion, and sedation. Treatment can also lead to excessive drying of the nasopharynx and thickening of the mucus deep in the throat. Botulinum toxin type A given every 3 months at a dose starting at 7.5 units into each parotid gland has been used effectively¹⁸⁸; however, serious side effects, including jaw dislocation, have been reported.¹⁸⁰ Radiation treatment is a fast, safe, inexpensive, and effective option but infrequently used.⁸ Side effects are mild and temporary. Surgical treatments for sialorrhea are available but not generally recommended.⁸

Respiratory Insufficiency. This is very rarely the presenting complaint for patients with ALS, though symptoms exist on presentation in around 25% of patients when specifically investigated. For the remaining majority, no respiratory symptoms are noted for some time after diagnosis. These symptoms can include fatigue, dyspnea, orthopnea, and morning headaches. Even though many patients have no respiratory symptoms, subclinical respiratory insufficiency is frequent. Up to 85% of patients with ALS will have an abnormal FVC at presentation.^{62,82} The majority of patients with less than 50% predicted FVC will have mild or no respiratory symptoms, even though NIV is both advantageous and covered by Medicare.⁸² For this reason, experts recommend routine assessment of respiratory physiologic parameters in patients with ALS at every visit, regardless of the presence of symptoms.

There are several different parameters that can be used to assess and monitor respiratory function. FVC is the most commonly utilized parameter.¹²⁸ The difference between the upright and supine FVC correlates best with orthopnea, which is the symptom most closely associated with quality of life.^{25,82} The maximal inspiratory pressure (MIP) decreases earlier than upright FVC.⁸² The sniff transdiaphragmatic pressure is most accurate at predicting hypercapnia (90% sensitive, 87% specific).¹²⁸ The sniff nasal pressure (SNP) is useful in a number of ways. It has greater predictive power and is more reliably recorded at later stages of disease than both FVC and MIP.¹²⁸ At a threshold of less than 32% predicted, SNP is 81% sensitive and 85% specific for predicting hypercarbic respiratory failure.⁸² In spite of this, FVC and MIP are the most established respiratory parameters and are currently used by most experts as well as Medicare to justify NIV. Unfortunately, no parameters can predict respiratory insufficiency in the many patients with ALS and bulbar dysfunction.^{8,115}

At some point, patients with ALS will need to decide if they wish to use respiratory support. Those who choose respiratory support usually begin with NIV. NIV is delivered through a mask or mouthpiece without the need for a tracheostomy. This can be

delivered with either a bilevel positive airway pressure machine or a portable ventilator connected to a mask. Both are portable and serve the same basic purpose. Portable ventilators have many more settings and can be used when and if the patient decides to undergo tracheostomy and invasive ventilation. Both improve respiratory function and quality of life.¹²⁸ Advantages to NIV include improved quality of life parameters, such as energy, vitality, dyspnea, somnolence, depression, sleep quality, physical fatigue, concentration problems, and cognitive function.^{128,142,173} It prolongs survival 205 days, though it is not clear if this applies to those with severe bulbar involvement.^{8,128} It also slows the rate of respiratory decline from -2.2% per month before treatment to -1.1% after initiation.¹²⁸ NIV delays the need for tracheostomy and mechanical ventilation but does not appear to add additional burden to caregivers.^{8,12,173} Most patients who have undergone NIV would choose it again and recommend it to other patients with ALS.^{8,12,128,173}

Thresholds for beginning NIV vary. Currently, Medicare requires an FVC of less than 50% or MIP less than -60 cm H₂O or PaCO₂ of 45 mm Hg or higher. The American College of Chest Physicians adds a criteria of nocturnal oximetry of 88% or less for 5 consecutive minutes.³ Early NIV, beginning with nocturnal oximetry with more than 15 desaturation events per hour, has been shown in nonrandomized trials to confer an 11-month survival benefit compared with controls using NIV at the thresholds listed earlier.¹²⁸ The setting for beginning use of NIV can vary. Use of NIV for greater than 4 hours per day offers a survival advantage of 7 months and a slower rate of respiratory decline when compared with those using it less.¹²⁸ A typical initial prescription for NIV would include the *International Classification of Diseases* (10th edition) G12.21 diagnosis for ALS and would include documentation of an FVC less than 50% or MIP less than -60 cm H₂O or PaCO₂ 45 mm Hg or higher.⁸²

In spite of the benefits of NIV, many patients are never offered it.⁸² About one-third of those who use NIV start it emergently.⁸² This is not ideal, but benefits of NIV can still be achieved when started emergently.⁸² Eventually, all patients will require more respiratory support and secretion management, and they will face the decision to undergo tracheostomy and invasive ventilation with a mechanical ventilator. This decision is often made urgently, when a patient is intolerant of NIV or unable to clear secretions.⁸² Triggers to consider invasive ventilation on a nonurgent basis include use of NIV longer than 12 hours per day or intolerance of NIV with an FVC less than 50% or symptoms of dyspnea.⁸² Tracheostomy and invasive ventilation can substantially prolong life; in one series, 47% of patients survived more than 5 years.¹⁷³ Much of this time was in a locked-in state where patients were unable to communicate.¹⁷³ Yet, patients report an 80% positive perception, would have it again, and would recommend it to another patient with ALS.⁸² Unlike NIV, there is a significant burden of invasive ventilation on caregivers, many of whom would not endorse invasive ventilation so strongly or at all. Only 50% of caregivers would consider it for themselves if they were to need it, and 30% of caregivers report a worse quality of life than the patient.⁸² Invasive ventilation via tracheostomy is costly as well, imposing an estimated \$150,000 or more per year on the health care system.

Diaphragmatic pacers have been shown to be of no benefit or may even decrease survival in patients with ALS.^{77,136,194} Oxygen therapy should almost never be used in patients with ALS. It worsens respiratory symptoms as well as hypercapnia and can lead to hypercapnic coma or respiratory arrest.^{13,82} Flow rates as low as

0.5 to 2 L/min can worsen CO₂ retention in patients with neuromuscular disease.⁸² The possible exception to this rule is patients in the terminal phase who may need oxygen for palliative relief of air hunger.

Bronchial secretions are often problematic and represent a major source of anxiety as well as being a negative prognostic factor for those on NIV.¹⁵⁶ First-line treatments include a portable home suction device and room humidifier. Mucolytics such as N-acetylcysteine, 200 to 400 mg three times daily, can also be helpful. Nebulized saline, beta antagonists, anticholinergic bronchodilators, and mucolytics have been used in various combinations with success. Surgical cricopharyngeal myotomy has been used to relieve cricopharyngeal spasm and severe bronchial secretions.⁸ High-frequency chest wall oscillation, used effectively in cystic fibrosis, has been shown not to work in two randomized trials, and the American Academy of Neurology indicates there are insufficient data to support or refute its use in patients with ALS.^{82,128} Breath stacking is a technique that involves multiple sequential inhalations before exhalation. It can increase inspiratory volume up to 20% and increase the peak cough flow (PCF) significantly.⁸² Patients can do this without assistance if they can close their glottis between inspiratory efforts; otherwise, a caregiver can use a resuscitation bag connected to a mask. A manual cough-assist can be performed by a caregiver. This consists of a caregiver leaning over a partially supine patient and applying downward pressure on the chest with one forearm, while thrusting on the abdomen with the other hand during expiration.⁸² The combination of manual cough assist and breath stacking is more effective than breath stacking alone.⁸² Manual cough assist should be avoided in patients with a Greenfield filter and used with caution in patients with recent abdominal surgery, abdominal feeding tubes, and those at high risk of rib fracture.⁸² If a patient has a vital capacity less than 340 mL or PCF less than 90 L/min, then transitioning to mechanical insufflation-exsufflation is typical.

A mechanical cough-assist device (mechanical insufflator-exsufflator [MI-E]) can be very effective via facemask and has been shown to be effective in uncontrolled trials.^{86,167} It works by providing positive airway pressure for a few seconds, then suddenly switching to negative pressure to simulate the pressures created in a healthy cough. Treatments usually consist of four to six insufflation-exsufflation cycles in a sequence. Sequences are separated by 30 seconds of rest. Four to six sequences make up a treatment, and patients usually receive three to four treatments per day.⁸² This can increase to one to two treatments per hour in the setting of a respiratory tract infection.⁸² The combination of NIV and MI-E has been shown to reduce hospitalization and improve oxygen saturation in patients with neuromuscular disorders, including ALS.⁸² A typical prescription would read⁸²:

Cough-assist device, 4 treatments per day and as needed, 5 coughing cycles per sequence followed by 30 seconds of rest, with 5 sequences for each treatment, beginning at +15 cm H₂O insufflation pressure/−15 cm H₂O exsufflation pressure, increased in 5–10 cm H₂O increments, to a target +40/−40 cm H₂O and oxygen saturation >94%.

Presently, this is covered by Medicare for patients with ALS and a PCF of less than 300 to 350 L/min (5–6 L/sec).⁸² It is less effective for patients with significant bulbar dysfunction, in whom a tracheostomy is often a better solution for secretion management. Contraindications include bullous emphysema, pneumothorax, and pneumomediastinum.⁸²

Mood Disorders. It is important for treating providers to discuss symptoms of mood disorders with patients (and families of patients) with ALS because treatment can lead to improved quality of life.¹¹⁴ Tricyclic antidepressants and selective serotonin reuptake inhibitors are often used. Tricyclic antidepressants have the added benefit of treating sialorrhea, pseudobulbar affect, and insomnia, but they are complicated by many side effects. Anxiety can be a particular problem when respiratory insufficiency develops, prompting use of benzodiazepines. They are effective but can suppress respiratory drive. Counseling is helpful to many patients and an integral part of most multidisciplinary clinics. Support groups through an organization such as the Muscular Dystrophy Association or groups specific to ALS, SMA, and other MNDs should be encouraged to patients who are open to the idea.

Pseudobulbar Affect. Pseudobulbar affect is pathologic weeping, laughing, and yawning that is inappropriate or excessive to the emotional state of the patient resulting from UMN dysfunction. It can consist of laughing or crying spontaneously when there is no apparent reason and can be alarming and distressing to those around the patient. The underlying mechanism is unknown but may be caused by loss of frontal lobe inhibition of spontaneous brainstem-generated emotional responses. Pseudobulbar affect is seen in over one-half of patients with ALS with or without bulbar motor signs.⁷⁰ It is not a mood disorder, does not reflect the emotional state of the patient, and can often be ignored. Caregivers are often more concerned than patients themselves but must be educated on the true nature of the expression. Multiple antidepressants and even levodopa and lithium have been used.¹⁷³ A combination of dextromethorphan and quinidine is US Food and Drug Administration (FDA) approved for treatment of pseudobulbar affect and has been shown to improve both symptoms and quality of life.²⁹ It is not clear how this combination helps pseudobulbar affect, but dextromethorphan is a known N-methyl-D-aspartate antagonist. About one-quarter of patients discontinued the combination, with common but mild side effects, including nausea, dizziness, and somnolence. There were increases in the QT interval seen in trials, but these were not clinically significant.²⁹

Cognition. Cognitive difficulties affect up to one-half of patients with ALS.⁷ Many of the gene mutations responsible for familial ALS can also cause FTD, and the reverse is also true, demonstrating an underlying link between the two processes. Approximately 15% of patients diagnosed with ALS will meet the criteria for FTD, and nearly one-third will meet criteria for mild cognitive or behavioral impairment.⁷ Symptoms of FTD may include apathy, change in emotional reactivity, sleep disturbances, stereotypical or repetitive behaviors, poor insight or judgment, and loss of expressive language with relative preservation of comprehension. Several cognitive screens have been developed to rapidly assess cognition in patients with ALS. Importantly, the Mini-Mental State Examination often will not detect mild cognitive or behavioral impairment in patients with ALS.⁷ In addition to complicating the care of patients who may already have other barriers to communication, frontotemporal syndrome is a negative prognostic factor for patients with ALS. Cognitive impairment can also complicate the ability to discern if a patient retains capacity to make end-of-life decisions.

Insomnia and Fatigue. Insomnia is a common symptom in ALS, particularly later in the disease. It is likely related to multiple factors, including pain, respiratory difficulties, mood disorders, and a disturbed sleep-wake cycle. The most commonly used medications include amitriptyline and zolpidem.¹⁹¹ Fatigue is another common and likely multifactorial problem affecting

many patients with ALS. Modafinil was helpful for ALS-related fatigue in one small class I study.⁷

General Medical Complications. Patients with ALS are at greater risk for venous thromboembolic events, presumably related to immobility and respiratory dysfunction. There have not been studies of pharmacologic prophylaxis, but limb elevation and compression stockings are recommended.⁷ Despite concerns, no convincing evidence has been shown to support an increased incidence of ALS or more rapid disease progression of ALS with the use of statins.⁹⁹ Patients with ALS do not require adjustments to vaccination guidelines or antibiotic use.⁹⁹ Surgery in patients with ALS is complicated because of many of the problems discussed earlier. A medical evaluation is recommended before surgery, with particular attention to the pulmonary system.⁹⁹ Regional anesthesia is generally recommended over general anesthesia.⁹⁹ Because of a higher incidence of post-operative respiratory depression, extubation should be withheld in patients with ALS until they are fully awake after the use of inhalation agents.⁹⁹ Depolarizing neuromuscular blockers are contraindicated in patients with ALS because of prolonged neuromuscular blockade.⁹⁹ NIV has been useful in patients with respiratory involvement after having received local anesthesia.⁹⁹

Prognosis and End-of-Life Care

End-of-life issues need to be addressed early in the management of every patient with ALS. Delaying uncomfortable end-of-life conversations may be tempting for both provider and patient alike. This may cause family members to make critical decisions during a crisis when they are unaware of the patient's true wishes. Establishing a health care proxy, living will, and Do Not Resuscitate orders helps avoid this. Clear and repeated communication of the patient's wishes to caregivers and family is still important even if all the necessary arrangements are in place. Palliative care specialists should be involved early.⁸ It is helpful to review the patient's wishes every 6 months to ensure that documentation is current and compatible with the patient's recent state of mind.

Providers often soften the blow of the initial diagnosis by making a promise: that everything will be done to keep the patient comfortable. The actual medical treatment of distressing symptoms in the last months of life is often done too conservatively. One survey found that in the last month of life, on a scale of 1 to 6 (6 representing constant suffering, 1 representing no suffering), caregivers gave an average rating of 4, and nearly half rated this final month as a 5 or 6.¹⁷³ It seems more needs to be done to deliver on this promise, and there may not be a more important task that a physician must undertake.

Common complaints in the last month of life are dyspnea, difficulty sleeping, choking episodes, severe pain, anxiety, confusion, and depression.¹⁷³ The initial management of many of these issues was previously discussed. Symptomatic treatment of intermittent dyspnea can consist of lorazepam 0.5 to 2.5 mg sublingually and morphine 2.5 mg orally or subcutaneously.⁸ Chronic dyspnea of a moderate degree can be treated with this same morphine regimen four to six times daily.⁸ Severe dyspnea can be treated with morphine intravenously or subcutaneously at 0.5 mg/hour and titrated to effect.⁸ Midazolam or diazepam can be used for nocturnal symptoms.⁸ Pain can be treated with non-narcotic medications but generally requires opioids, which have many cross benefits for dyspnea and anxiety. Benzodiazepines are usually used for anxiety as well as dyspnea.

Providers are often concerned about overtreating the patient, but palliative care experts assert that titrating doses of benzodiazepines

• BOX 41.5 Important Statistics in the Prognosis of Amyotrophic Lateral Sclerosis

Average survival from onset of symptoms: 2–3 years

Average survival from diagnosis: 15 months

3-year survival rate (from diagnosis): 41%

5-year survival rate (from diagnosis): 20%

Factors negatively impacting prognosis at diagnosis:

- Female sex

- Increasing age

- Bulbar onset

- Rapid early progression

- More definite diagnostic category

Treatment factors impacting prognosis:

- Percutaneous endoscopic gastrostomy

- Noninvasive ventilation

- Riluzole

Participation in multidisciplinary clinics

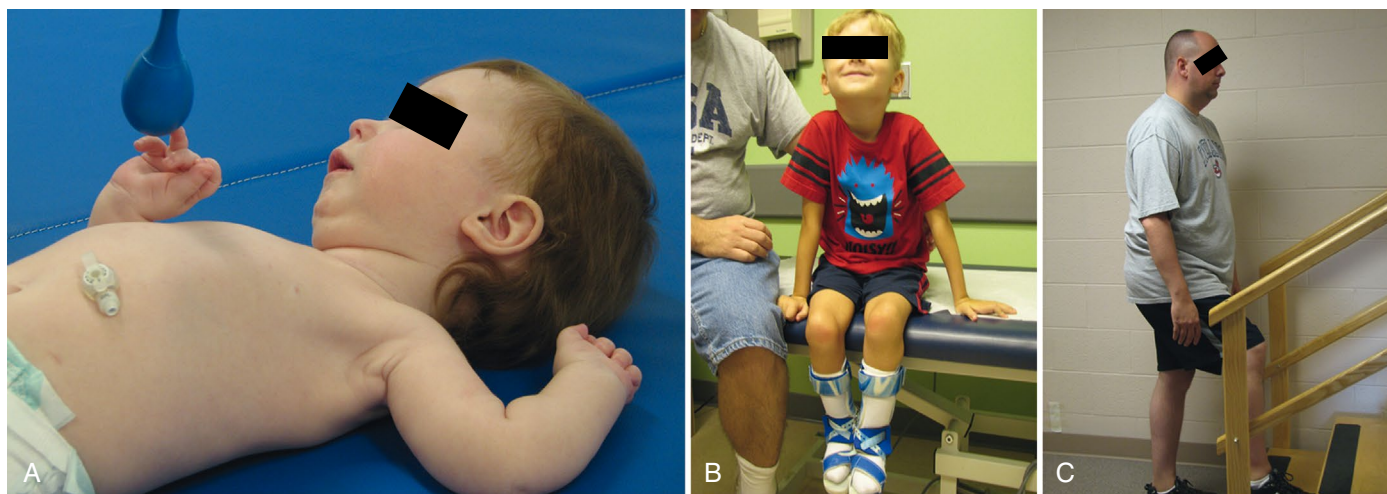
and opioids to clinical symptoms will almost never result in life-threatening respiratory depression.⁸ Supplemental oxygen, however, can suppress respiratory drive and cause hypercarbic respiratory failure. It should only be given to patients with symptomatic hypoxia. Restlessness and confusion caused by hypercapnia can be treated with neuroleptics such as chlorpromazine 12.5 mg every 4 to 12 hours orally, intravenously, or as needed. Patients and caregivers can be reassured that death itself is peaceful. Surveys have found that 91% of deaths caused by ALS in the United States and Canada were considered peaceful by caregivers.¹⁷³

The prognosis for ALS is grim (Box 41.5). There is a small (10%) subset of patients with longevity for reasons that are unclear. A fortunate half of these have a more benign course; the other half have prolonged but severe involvement.¹⁴⁴ Overall survival from the onset of symptoms averages between 2 and 3 years, and survival after the diagnosis is made is 15 months on average, with significant variability between individual patients.^{39,83,144} The prognosis is worse with age and bulbar involvement. Males survive longer than females possibly because of their lower incidence of bulbar involvement.³³ Malnutrition causes a nearly eightfold increase in the risk of death.⁵³ Those with more diffuse involvement and a more rapid rate of decline have a shorter survival. PEG tube placement, NIV, treatment with riluzole, and participation in multidisciplinary clinics improve survival.^{40,182}

Spinal Muscular Atrophy

General

SMA is a genotypically and phenotypically diverse group of LMN disorders. The most common and well-understood form, proximal SMA (also called SMN-related SMA, 5q-SMA, or simply SMA), is an autosomal recessive lower MND with a frequency of 1/11,000 births.^{154,161} Carrier frequency is about 1/50.¹⁷⁸ Importantly, untreated proximal SMA is the most common genetic cause of death in infants.¹⁶⁴ Proximal SMA historically has been classified into five subtypes of disease, though expected disease onset and progression has changed with availability of new therapies (Fig. 41.4).^{9,58,157,199} The most severe form, type 0, has onset before birth, with very severe weakness and often joint contractures caused by diminished intrauterine movement (not shown in Fig. 41.4). Type 1 is the most common form of SMA, representing 60% to 70% of cases; it is



• **Fig. 41.4** Subtypes of proximal spinal muscular atrophy (SMA) related to *SMN1* deletion. (A) SMA type 1, the most common form, has onset before 6 months of age, and the ability to sit upright is never achieved without treatment. (B) SMA type 2 has onset between 6 and 18 months of age; the ability to sit independently is achieved but standing is not without treatment. SMA type 3 is associated with onset after 18 months and patients are able to stand or walk at least temporarily. (C) SMA type 4 is the mildest subtype with onset after 30 years of age. Type 0 with onset before birth is not shown.

associated with onset before 6 months of age and an inability to sit independently without treatment. Historically, 95% of type 1 SMA patients die by the age of 2 years without treatment.²³ Patients with type 2 have onset between 6 and 18 months of age; they achieve the ability to sit upright independently but are unable to ambulate without treatment. Patients with type 3 have an onset after 18 months of age and are able to walk independently at least temporarily. The mildest form, type 4, is marked by onset in adulthood with relatively mild proximal limb weakness.¹⁵⁷

SMA is caused by homozygous deletions or loss-of-function mutations of the *SMN1* gene, which is located on chromosome 5q13.¹¹³ In ~95% of cases, patients have a homozygous deletion of the *SMN1* gene, but 5% of patients will be compound heterozygotes with deletion of one *SMN1* allele and a missense mutation of the other *SMN1* allele. A second, closely related gene, *SMN2*, is retained in SMA patients in varying copy numbers. *SMN1* and *SMN2* genes encode the same SMN protein, but because of a single nucleotide C-T transition in exon 7, the *SMN2* gene in 90% of transcripts produces a shortened, unstable form of SMN that does not function normally and is rapidly degraded. In SMA patients, there is homozygous loss of both *SMN1* alleles, and reduced amounts of full-length SMN protein are produced by retained copies of the *SMN2* gene.⁹ Consequently, SMA is not related to absence of the SMN protein but, rather, reduced levels. Given the varying copy numbers of the *SMN2* gene in the general population, there are variable levels of full-length SMN protein production. Generally, SMA type 1 is associated with two copies of *SMN2*. In cross-sectional studies, the copy number generally correlates inversely with severity of SMA; however, it is important to note that *SMN2* copy number cannot precisely predict severity in an individual patient.^{31,61,118,192} The mechanism of how motor neuron dysfunction is related to reduced SMN levels remains unknown.⁹

SMA Assessment

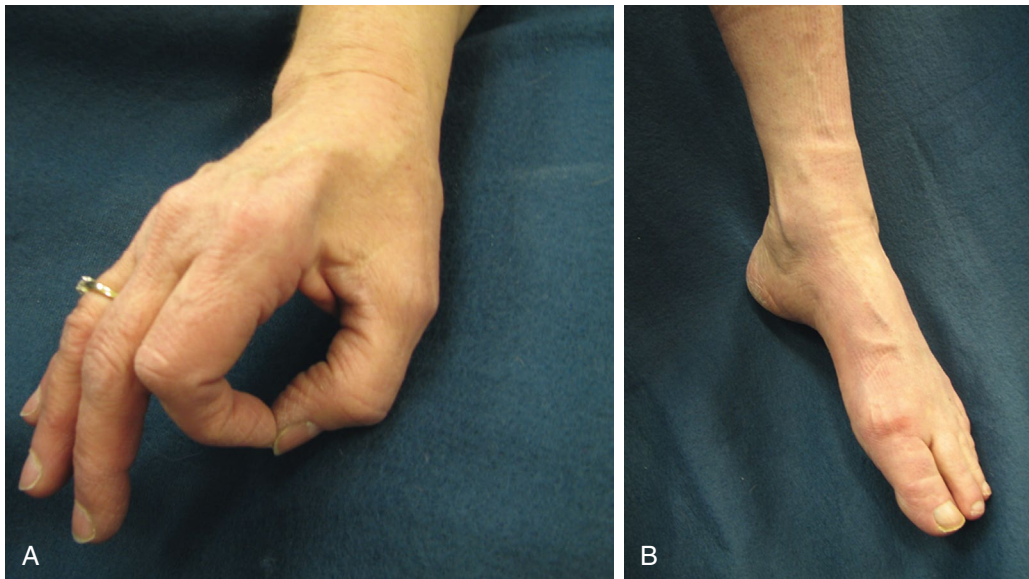
Clinical presentation. The clinical presentation of SMA includes predominantly proximal weakness and hypotonia. Reflexes

may be absent or reduced. Sensory examination is typically normal. About 60% to 70% of cases of SMA will have onset of weakness and hypotonia after birth but before 6 months of age and will not gain the ability to sit independently; these cases are classified as type 1, but in milder forms of SMA, symptoms may not be noted until adulthood. Other characteristic clinical features may include a fine tremor (polyminimyoclonus), tongue fasciculations, and paradoxical abdominal movement caused by relative sparing of diaphragm strength. The diagnosis of proximal SMA should be suspected in any infant developing hypotonia and weakness.

Laboratory studies. The most efficient diagnostic strategy is genetic testing for homozygous deletion of the *SMN1* gene, which can identify 95% of proximal SMA cases. Yet 5% of SMA patients are compound heterozygotes with a single deletion and a missense mutation. In such cases, *SMN1* deletion testing is normal. Subsequent analysis with *SMN1* dosage testing to identify a single *SMN1* deletion and sequencing the remaining *SMN1* gene are required to confirm the diagnosis of SMA.

Other testing. Strategie include electrodiagnosis and muscle biopsy, but given the availability of molecular diagnostics, such strategies are reserved for atypical cases or SMA not related to the *SMN1* gene. When performed, electrodiagnosis correlates with clinical severity and age and shows variable features of motor neuron/axon loss consistent with loss of motor neuron function.^{30,87,179} Sensory studies are usually normal.^{30,87,179}

Other gene mutations have been associated with autosomal recessive forms of proximal SMA, including *IGHMP2*, which is associated with SMA with respiratory distress, and *Gle1*, which is associated with severe lethal congenital contracture syndrome 1 and ultimately fetal loss of motor neurons.^{81,190} Autosomal dominant forms of proximal SMA have been associated with mutations in *VAPB*, *TRPV4*, *LMNA*, and recently *BICD2*.^{45,141,147,155} X-linked recessive SMA is associated with *Ube1* mutations.¹⁸ Some forms of SMA are associated with a distal predominant pattern of weakness and therefore have been described with the term distal SMA (Fig. 41.5). However, because there are notable phenotypic and genotypic overlaps between distal SMA and other



• **Fig. 41.5** Distal predominant upper limb (A) and lower limb (B) atrophy in a patient with distal hereditary motor neuropathy mimicking that of Charcot-Marie-Tooth disease.

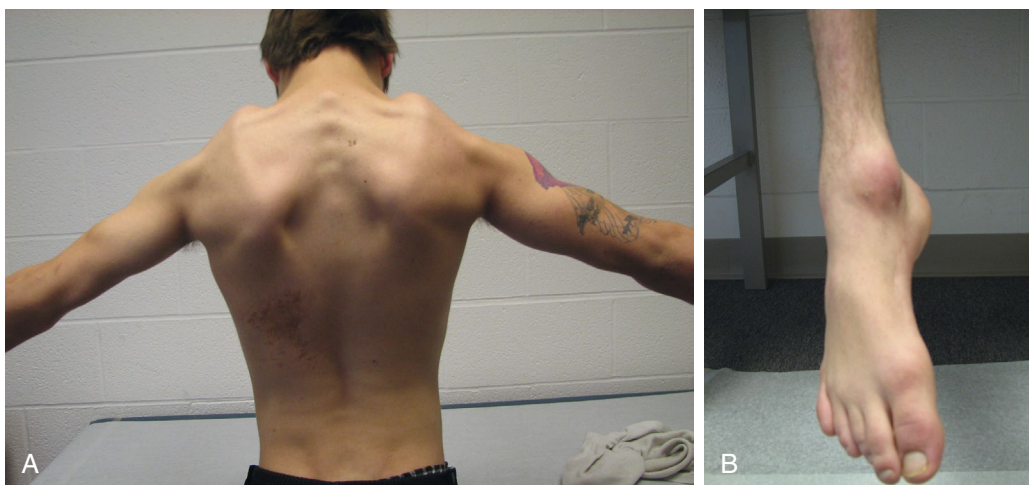
hereditary neuropathies (i.e. Charcot-Marie-Tooth disease), the term distal hereditary motor neuropathy (dHMN) is more frequently used. An ever-expanding number of genetic mutations have been associated with dHMN, including GARS, DCTN1, HSPB8, HSPB1, BSCL2, SETX, HSPB3, DYNC1H1, REEP1, and SLC5A7 mutations.^{45,90} dHMN presents with clinical features of distal weakness, atrophy, and reduced to absent reflexes. Sensory function, by definition, is preserved, but some patients will have features of sensory loss closely mimicking axonal forms of Charcot-Marie-Tooth (see Fig. 41.5). A rarer form of SMA, called scapuloperoneal SMA or Davidenkow syndrome, is marked by features of motor neuron and axonal loss in a periscapular and distal leg distribution, mimicking the pattern of facioscapulohumeral muscular dystrophy (Fig. 41.6). This syndrome has been linked to mutations in the *TRPV4* gene.^{94,186} Chronic or adult-onset hexosaminidase deficiency, sometimes called late-onset Tay-Sachs, can cause a clinical syndrome of motor neuron loss similar

to that of proximal SMA and is sometimes associated with features of cerebellar degeneration, dystonia, and psychosis.¹⁵²

SMA Treatment

Disease-modifying medications. There are currently three genetic-based treatments for SMA: nusinersen, onasemnogene abeparvovec, and risdiplam.¹⁶²

Nusinersen. This medication was the first FDA-approved therapy for all types of SMA in both pediatric and adult patients. It is an antisense oligonucleotide that corrects *SMN2* mRNA splicing, leading to the production of functional SMN protein.^{92,176} Because of its large size and inability to cross the blood-brain barrier, nusinersen is administered intrathecally. The recommended dosage is 12 mg per administration, starting with three loading doses given at 14-day intervals and a fourth after a 30-day interval. Maintenance doses are administered every 4 months. The most common side effects include lower respiratory infection, fever,



• **Fig. 41.6** A patient with the unusual pattern of periscapular (A) and distal leg muscle atrophy (B) associated with the scapuloperoneal form of spinal muscular atrophy (Davidenkow syndrome) with striking similarity to that seen in facioscapulohumeral muscular dystrophy.

constipation, headache, vomiting, back pain, and postlumbal puncture syndrome.¹³⁹ In related clinical trials, nusinersen has demonstrated the necessary safety requirements along with significant improvements in achieving motor milestones and stabilizing motor function. Overall, data from trials suggest that earlier treatment initiation leads to better stabilization of motor function. There are still gaps in the research on nusinersen. The long-term efficacy of nusinersen into adulthood is yet to be demonstrated.⁵⁹

Onasemnogene Apeparovvec. This gene replacement therapy offers a single intravenous treatment, contrasting with the frequent and invasive dosing of nusinersen. It uses adeno-associated virus 9, a blood-brain barrier-penetrant virus, to deliver *SMN1* cDNA under a chicken beta-actin hybrid promoter to all tissues via the circulatory system.⁶⁶ Trials of onasemnogene apearovvec found that among 12 patients who received the higher dose, 11 could sit unassisted, speak, and feed orally, and two could walk by 20 months of age.¹²³ The higher dose became the therapeutic standard, with all patients surviving without permanent ventilation and maintaining motor milestones for 5 years.¹²⁵ However, significant liver toxicity was noted, with elevated serum aminotransferases managed by prednisolone treatment.⁸⁴ Onasemnogene apearovvec is now approved by the FDA for patients under 2 years old with biallelic mutations in the *SMN1* gene.⁸⁴ Despite elevated liver enzymes in over one-half of the patients, the one-time intravenous dose of onasemnogene apearovvec significantly improved the achievement of key developmental motor milestones.^{34,50} Data are still needed to fully understand the long-term effectiveness of this therapy.

Risdiplam. This daily oral medication is FDA approved for treating SMA patients age 2 months and older. It provides a less invasive alternative to nusinersen and onasemnogene apearovvec.⁴⁷ As an *SMN2* pre-mRNA splicing modifier, risdiplam increases SMN protein production and distributes throughout the central nervous system and peripheral organs, increasing SMN protein levels body-wide.¹⁵⁸ Risdiplam should be taken orally once daily after a meal, at the same time each day. Results from a phase 2/3 study on infants with SMA type 1 showed improved motor function, with 29% able to sit unassisted for 5 seconds after 12 months.⁴⁶ Another phase 2/3 study evaluated children and young adults with type 2 or 3 SMA. By 12 months, patients treated with risdiplam had a Motor Function Measure 32 score 1.55 points higher than the placebo group, increasing to 1.83 points by 24 months.¹⁶ The treatment showed less improvement over time in older individuals but was more effective in stabilizing symptoms in these patients. Common adverse reactions include fever, diarrhea, rash, mouth ulcers, arthralgia, urinary tract infections, upper respiratory infection, pneumonia, constipation, and vomiting. Risdiplam is contraindicated for patients with hepatic abnormalities.^{4,54} Additionally, risdiplam may impact male fertility, which is particularly concerning for adult male patients.^{151,162}

Rehabilitation, symptomatic and medical treatment. SMN-restoring treatments have altered the treatment landscape for patients with SMA and physiatrists treating patients with SMA. Earlier implementation of SMN-restoring treatments clearly demonstrates more pronounced improvements in motor function.^{42,124} Improved strategies are needed for earlier diagnosis and treatment, including genetic screening, to improve outcomes. More work is needed to optimize SMN-related therapeutics, to understand the long-term impact of genetic therapies, and to understand future needs of patients affected by SMA.

Despite the approval and availability of SMN-restoring treatment, most patients continue to require some aspect of medical

and rehabilitation management to address musculoskeletal, pulmonary, nutritional, and gastrointestinal complications. Restrictive lung disease is a major issue that should be aggressively managed similar to other MNDs. Pulmonary disease is the main source of mortality; muscle weakness can cause complications related to impaired ventilation and secretion management, as well as secondary complications such as aspiration pneumonia. Pulmonary complications are related to the severity of disease, wherein greater disease burden requires closer monitoring and more frequent intervention. Establishing a therapeutic relationship with an experienced pulmonary specialist familiar with the management of SMA is critical and should occur at the time of initial diagnosis to anticipate future needs and prevent complications. A multidisciplinary team approach that incorporates the patient's and family's wishes, as well as the patient's best interests, should guide the level of ventilatory support offered.

Bulbar weakness is common in severe forms of SMA, predisposing to inadequate nutritional intake. Therefore, nutritional support is an important aspect of supportive care, with goals of avoiding under or over nutritional supplementation and reducing the risk of aspiration pneumonia. Scoliosis is common in SMA, and its progression and the associated impact on pulmonary function are less severe in patients with SMA type 3 compared with type 2. A corset brace does not halt scoliosis progression but may help provide seating stability and delay surgical correction. Surgical treatment is well established in nonambulatory patients at age 10 to 12 years with progressive scoliosis.

X-Linked Spinobulbar Muscular Atrophy (Kennedy Disease)

KD (or X-linked spinobulbar muscular atrophy) is an X-linked recessive disorder that leads to progressive limb and bulbar weakness, testicular atrophy, gynecomastia, muscle cramps, and fasciculations.¹⁰⁵ KD is related to an expanded CAG trinucleotide repeat in the q arm of the X chromosome within the first exon of the androgen receptor gene.¹¹⁰ Healthy individuals have 10 to 36 repeats in this region, whereas patients with KD have 40 to 62 repeats. Usually female carriers are unaffected, but some investigations have suggested subclinical features in manifesting female carriers, including muscle cramps and tremor. Electrodiagnostic or muscle biopsy evaluation may show features consistent with denervation.⁸⁰ Early manifestations of the disease in affected males include symptoms of muscle cramps, fasciculations, and tremor; whereas progressive muscle loss, weakness, and aspiration pneumonia from bulbar involvement are evident later in the disease course. KD typically has an onset of more overt symptoms by the fourth or fifth decade. KD is a slowly progressive disorder, and therefore lifespan is not usually dramatically reduced.

Clinical examination demonstrates proximal predominant and bulbar muscular weakness and atrophy; however, distal involvement and asymmetry can occur. This variability can sometimes complicate the clinical evaluation and more closely mimic the presentation of ALS. Key differences from ALS include absent or reduced reflexes, a lack of other UMN signs, often mild asymptomatic sensory loss, and in some patients features of androgen insensitivity (e.g., gynecomastia, reduced fertility). In addition, patients with KD often have prominent bulbar involvement with profound atrophy and perioral fasciculations (Fig. 41.7).

Genetic testing for CAG repeats in the androgen receptor gene is diagnostic of KD, but because of a significant overlap with other MNDs, electrodiagnostic testing is often obtained. EMG



• **Fig. 41.7** Typical features of atrophy seen in bulbar and tongue muscles in a patient with Kennedy disease.

shows features of active and chronic denervation with fibrillation potentials, fasciculation potentials, and enlarged MUAPs along with reduced recruitment in a generalized distribution.⁶⁴ Facial muscle investigation usually shows features of fasciculations and myokymic potentials as well as fibrillation potentials and neurogenic MUAP changes. NCS may show reduced compound muscle action potential amplitudes and relatively preserved conduction velocities. A distinctive feature of KD is a superimposed sensory neuropathy.⁶⁴ Importantly, this neuropathy is non-length-dependent in distribution, which is in contrast to the typical pattern of neuropathy that may be occasionally seen incidentally in patients with ALS. The electrodiagnostic findings of non-length-dependent sensory neuropathy is consistent with the fact that KD is associated with both sensory and motor neuropathy, but the features of LMN involvement in KD are more clinically evident and relevant. CK is moderately elevated in the majority of patients with KD to levels between 5 and 15 times that of normal.⁶⁸

There is no curative treatment for KD. Mouse models of KD have shown improvement after treatment with androgen receptor antagonists.⁹⁶ Supportive treatment is similar to ALS without the need to treat complications of UMN dysfunction, such as spasticity and pseudobulbar affect. Respiratory and secretion management is particularly important given the high incidence of bulbar dysfunction.

Poliomyelitis, Postpolio Syndrome, and Acute Flaccid Myelitis

Poliomyelitis

Poliomyelitis is a MND caused by viral infection of the central nervous system resulting in the loss of anterior horn cells and cranial nerve nuclei. Poliomyelitis results primarily from infections with the poliovirus, a human enterovirus of the Picornaviridae family. Other enteroviruses and flaviviruses, such as West Nile virus, are also associated with the clinical syndrome of poliomyelitis, but the poliovirus itself is the most widespread and well-known cause.^{91,168} There are three serotypes of the poliovirus, but immunity is not conferred between serotypes as a result of differing capsid proteins and antigenicity.¹³³ There has been a dramatic

reduction in the annual number of cases of polio as a result of the development of vaccination strategies in the 1950s, and in 1988, the World Health Assembly formulated plans for worldwide efforts to eradicate polio. These efforts have led to a dramatic reduction from over 300,000 cases worldwide to only 12 cases of wild-type polio reported in 2023, and those were restricted to Pakistan and Afghanistan.² Circulating vaccine-derived poliovirus can occur in populations of underimmunized children from excretion of oral live poliovirus that then mutates and becomes pathogenic. Importantly, fewer than 800 cases of vaccine-derived poliomyelitis have occurred over a period that would have incurred an estimated 6.5 million cases of wild-type polio without the vaccine.²

After individuals are exposed to the virus via the fecal-oral route, the poliovirus has a variable incubation period, ranging from a few days to over a month. During this period, the virus replicates in the gastrointestinal tract. There is a wide spectrum of disease severity, ranging from asymptomatic infection to severe paralysis and death. Before the availability of vaccination, polio represented the most common cause of acute flaccid paralysis. Individuals with poliovirus infection are often asymptomatic but may have symptoms of headache, fever, stiffness, and pain.⁹¹ Less than 5% of infections result in irreversible paralysis, and in these affected individuals, mortality can be as high as 10% from autonomic dysfunction, circulatory collapse, and ventilatory muscle failure.⁹¹

Symptoms of asymmetric flaccid paralysis and subsequent atrophy are more common in limb muscles but may also affect bulbar muscles. Rarely, transverse myelitis associated with features of autonomic, sensory, and sphincter dysfunction can occur. The clinical presentation of poliomyelitis can closely resemble other disorders affecting the spinal cord, muscles, and nerves.⁹¹ Therefore, when diagnosing, it is crucial to consider and rule out other potential causes of acute flaccid paralysis. Electrodiagnostic testing shows features of asymmetric or multifocal denervation. CSF analysis demonstrates elevated protein levels, but in contrast to other common causes of acute flaccid paralysis such as Guillain-Barré syndrome (GBS), there is evidence of pleocytosis. MRI can show features of increased T2 signal intensity in the region of the ventral horn of the spinal cord.¹¹⁹ Molecular testing with polymerase chain reaction can confirm viral serotype and can determine whether the infection is vaccine derived or is a wild-type strain. Management of acute poliomyelitis is supportive and is similar to other disorders of flaccid paralysis.

Postpolio Syndrome

After paralytic poliomyelitis, the surviving motor neurons can compensate and, through collateral reinnervation, expand their territories to provide some functional recovery.⁴⁴ This phase usually maximizes within 2 years of disease; thereafter, motor system function stabilizes. For yet-to-be determined reasons, approximately 50% of patients with poliomyelitis develop late features of declining motor function, usually 30 or more years after onset, with onset that may be gradual or abrupt.⁴⁴ A thorough evaluation is critical to exclude other disorders that may present as a decline in function. Treatment of postpolio syndrome is designed to maximize and maintain function. Therapeutic rehabilitation typically includes strategies to allow for energy conservation by both modifying the environment and delivering appropriate equipment needs similar to other disorders of the motor neuron.

Acute Flaccid Myelitis

Acute flaccid myelitis (AFM) is a syndrome first noted in the United States in 2014, seen primarily in children and thought to

result from infection from enteroviruses D68 and A71, both from the same family as the poliovirus. The neurologic syndrome is typically preceded by nonspecific viral symptoms and consists of patchy acute flaccid limb paralysis that evolves over 48 to 96 hours.¹⁴⁰ Cranial nerve involvement is seen in about 25% of patients.⁷⁶ Sensory findings are minimal, and in this way AFM resembles poliomyelitis as well as the acute motor axonal neuropathy variant of GBS. The weakness seen in AFM is usually in a classically patchy distribution, unlike the ascending pattern often seen in GBS, and the weakness of AFM tends to progress more rapidly. CSF examination shows cytoalbuminologic dissociation similar to GBS. The MRI findings of increased T2 signal in the anterior horns is the most helpful test finding to distinguish AFM from GBS. Treatment usually consists of high-dose corticosteroids, intravenous immunoglobulin, or plasma exchange. Maximum improvement is usually seen by 6 months, but a full recovery is seen in less than 20% of patients; close to 15% require long-term use of assistive devices for ambulation or are completely dependent on caregiver support.⁷⁶

Rare and Less Well-Defined Etiologies of MND

Beyond the more common forms of MND, a myriad of other unusual and atypical forms are less well-defined and preclude a detailed description within this chapter. Some of the uncommon forms have been attributed to paraneoplastic and idiopathic immune dysregulation, other infectious agents, hereditary metabolic disorders, electrical injuries, and other idiopathic processes. Fortunately, most of these are exceedingly uncommon. Because of the sheer rarity of MNDs as a whole, investigation into cause and effect vs. chance association is difficult. Search of the literature often reveals confusing data in this regard. One such association is that of disorders of the endocrine system and MND. Primary hyperparathyroidism may lead to a condition of MND, but in regard to the pathogenesis of ALS, most data suggest this association is coincidental.⁹⁵

Hirayama Disease

HD is a self-limited disorder associated with muscle weakness and atrophy of the distal upper limb muscles. Several names have been used to describe HD, including monomelic amyotrophy, distal muscular atrophy of the distal upper extremity, and oblique amyotrophy. HD was first described in 1959 by Hirayama.⁸⁹ The

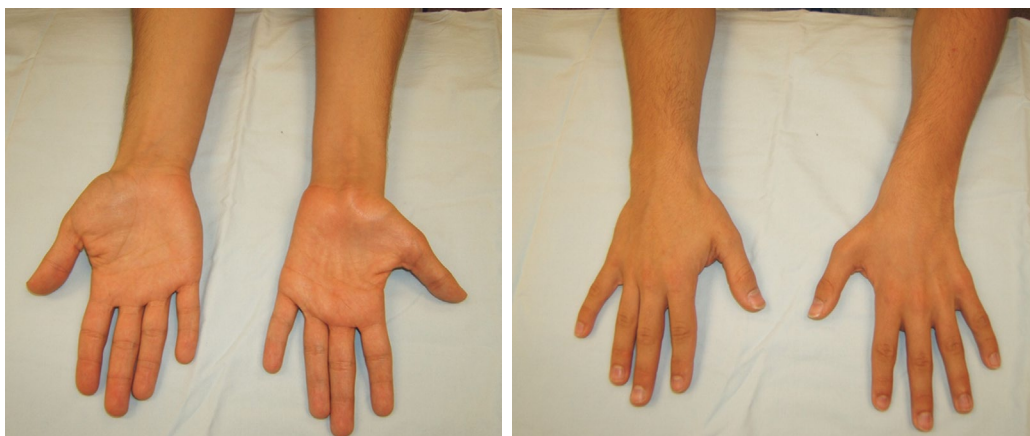
pathogenesis of HD has been attributed to the anterior displacement of the dural wall causing impaired microcirculation of the lower cervical spinal cord and loss of motor neurons.^{41,60} The disorder primarily affects males during the teenage years or early 20s, and the onset of weakness is insidious and slowly progressive. The natural history of the disease is associated with progression for 5 years or less.^{75,78} Involvement may be unilateral or bilateral but is usually more prominent in one limb (Fig. 41.8). Interestingly, preponderance for right-sided involvement has been noted.⁷⁸ The phenomenon of worsening weakness with exposure to cold, termed cold paresis, has been frequently described in patients with HD.⁷⁸ Features of UMN signs, cranial nerve abnormalities, or incontinence of bowel or bladder are not seen, and sensory disturbance is usually absent or minimal.

Electrodiagnostic assessment demonstrates active and chronic denervation in muscles innervated by the C7-T1 myotomes with features of fibrillation potentials and enlarged MUAPs with decreased recruitment. Fasciculations are not apparent on clinical or electrodiagnostic examinations. MRI with neck flexion is used to assess for forward displacement of the posterior dural sac.^{38,60,106}

Because of the insidious onset, slow progression, and relatively uncommon nature of the disorder, there is typically significant delay in recognizing HD.¹¹⁶ Although HD is a self-limited disorder, early diagnosis is critical to avoid delay in treatment and avoid unnecessary loss of upper limb function.¹¹⁶ Treatment may include using a cervical collar to limit neck flexion. In patients with continued progression despite conservative management, cervical spine surgery for decompression and stabilization may be indicated.^{116,181} Early treatment can demonstrate significant efficacy, which supports the belief that the origin is related to structural dysfunction rather than a true degenerative process.^{75,116}

Paraneoplastic Degeneration

Paraneoplastic degeneration of the UMNs and LMNs has been uncommonly reported as a cause of MND. In most cases reported, the link between MND and paraneoplastic degeneration is purely that of coincidental association. In some cases, an etiologic link has been suspected on the basis of a temporal relationship between the onset of MND and cancer, the presence of autoantibodies, and the response to immunomodulatory treatments such as intravenous immunoglobulin or corticosteroids. The causative relationship between cancer and the development of motor neuron degeneration is felt to be unlikely in most cases of



• Fig. 41.8 Asymmetric left hand and forearm muscle atrophy in a patient with Hirayama disease.

cancer and MND.⁶⁵ Some authors have even suggested that paraneoplastic processes could unmask FALS or increase the penetrance underlying FALS-related gene mutations.^{55,72} One of the best defined paraneoplastic syndromes associated with MND is an anti-Hu syndrome.⁶⁵ The clinical spectrum of anti-Hu-related paraneoplastic disorders includes sensory neuropathy, cerebellar ataxia, and limbic encephalitis; this overlaps with syndromes of multifocal involvement and syndromes of motor neuron or motor axonal loss, often with sensory involvement.⁷⁹ The presence of an anti-Hu MND-like phenotype is infrequent at less than 5% of cases with anti-Hu syndromes.⁷⁹ Anti-Hu antibodies are not felt to be pathogenic but appear to be a marker of autoimmunity against Hu antigens.⁷⁹ Treatment includes management of the underlying neoplasm and in some cases immunomodulation.

Nonpoliomyelitis Infections

Infections other than poliomyelitis have been more rarely implicated in the pathogenesis of MND.^{85,143,166} Of these infections, the rare association of human immunodeficiency virus (HIV) infection with MND is well established.^{6,117} HIV infection can involve either UMNs or LMNs, and patients predominantly present with mixed UMN and LMN features. The disease may present at varying stages of HIV infection. A clinical pattern of progressive motor neuron degeneration over weeks rather than months along with pathologic features of vacuolar myelopathy and inflammation suggest an immune pathogenesis that is distinct from ALS.¹⁶⁶ Recognizing this syndrome is important because at least half of cases will respond to adequate antiretroviral therapy, further supporting a link to HIV infection and pathogenesis.^{6,148} Other infectious agents implicated in underlying MND, including Lyme disease (*Borrelia burgdorferi*), herpes zoster, Creutzfeldt-Jacob disease, and human T-lymphotropic virus-1 (HTLV-1), are more controversial. HTLV-1 infections are typically associated with an UMN-predominant syndrome of myelopathy called tropical spastic paraparesis or HTLV-1-associated myelopathy (HAM).^{74,197} This disorder is distinct from MND because of the presence of symmetric lower limb involvement, sensory loss, and bowel and bladder changes. HAM occurs in about 1% to 3% of individuals infected with HTLV-1.¹⁹⁷ HTLV-1 infection is endemic in certain areas, and therefore serologic evidence of prior infection should not be used to prove a causative relationship. Herpes zoster can result in segmental weakness and atrophy with features that overlap with MND. It is not usually considered to be an isolated disorder of the motor neuron but affects the root, plexus, nerve, brain, and rarely the spinal cord.

Electrical Injury

There are sparse reports of electrical injury preceding syndromes of motor neuron loss.⁵ The clinical features of such disorders are associated with variable delays of days to years after electrical injury. The epicenter of motor neuron degeneration loss occurs at the site of the electrical entry or exit in 90% of cases.⁵ Myelopathic features can also be present in some cases, including sensory symptoms, pain, and bowel and bladder dysfunction.

Hereditary Bulbar Syndromes

Brown-Vialetto-Van Laere syndrome-1 (BVVLS-1) and Fazio-Londe disease (FLD) are two closely related and extremely rare childhood syndromes caused by autosomal recessive mutations in the *SLC52A1*, *SLC52A2*, and *SLC52A3* genes, which encode riboflavin transport proteins.²⁴ BVVLS-1 and FLD both cause progressive motor neuron loss with prominent bulbar and respiratory

involvement. Usually BVVLS-1 is distinguished from FLD by the universal presence of hearing loss, but the two syndromes are now usually considered synonymous. Onset of both syndromes typically occurs in the first 2 decades of life. The pathogenesis of both syndromes is related to riboflavin deficiency; treatment therefore includes high-dose riboflavin supplementation.

General Rehabilitation, Symptomatic and Medical Treatment of MND

Most MNDs have a profound impact on the overall life and function of affected patients, and a comprehensive program of rehabilitation with appropriate symptomatic and medical treatment is required to optimize patients' function and quality of life. This is usually best achieved through a multidisciplinary approach, whether through a formal multidisciplinary clinic or the coordination of the various disciplines by the physiatrist treating the patient.

Exercise in MND

In general, exercise has been shown to be safe, effective, and well tolerated in MNDs. Yet, the role of exercise in managing MND is the subject of controversy, and research is lacking. Specifically in ALS, retrospective epidemiologic data have suggested increased MND, specifically ALS, attributable to increased participation in strenuous exercise.⁹⁸ Despite theoretic concerns of worsening weakness, no controlled outcome study has shown worsening after exercise in patients, and broadly, studies show benefits of moderate aerobic and resistance exercise in MND.^{43,126,149} For ALS, which is a progressive disease, noncontrolled trials cannot demonstrate a greater decline in strength or function than what would naturally occur over time. A 2013 Cochrane review found limited evidence of benefit but no evidence of harm of exercise, and a 2020 meta-analysis concluded that long-term functional improvements but no change in quality of life or harm could be attributed to exercise.¹²⁶ Overall, at this point in time there are insufficient data to establish the benefit (or harm) of exercise in patients with ALS with a high degree of certainty, but functional benefits are likely, and the risk of harm is low.⁴³

ROM and stretching are considered safe and should be prescribed for all patients with ALS. This is especially helpful in preventing contractures that could be a source of pain later in the disease.⁴⁹ Using the data available, it is reasonable to have patients or caregivers apply ROM and stretching exercises to all patients with ALS and to encourage more functional patients to engage in mild aerobic and resistance training. The latter can consist of mild to modest resistance exercises for 15 minutes twice daily. The potential for overwork weakness can be avoided by some commonsense precautions, such as avoiding resistance exercise to fatigue or cramping, aerobic exercising to the point of significant dyspnea, eccentric exercises, or exercising muscles with significant weakness. An unusual amount of delayed-onset muscle soreness is a sign to decrease the intensity of the program going forward. Care must be taken to avoid falls and injuries. Stationary cycling or pool therapy can reduce this risk. Overall, while a definitive evidential basis is lacking, exercise remains one of the few modalities that can improve the function of a patient with ALS. Some form of exercise should be discussed with every patient with ALS. In addition to the benefits in function a reasonable exercise program can offer, it allows the patient to exert some control over the disease in an active manner.

The effects of exercise in other MNDs are similarly hampered by a lack of data. A 2019 Cochrane review could draw no definitive conclusions regarding the benefit or harm of exercise in SMA,¹⁷ nor could a 2021 Cochrane review of exercise in KD,⁹⁷ though one patient with KD showed a subjective functional loss and increased CK after exercise.¹⁵⁹ European Federation of the Neurological Societies guidelines from 2006 indicate good evidence for resistance and aerobic exercise in patients with PPS,⁶³ but a Cochrane review including more recent trials could conclude only that there was very low-quality evidence that muscle strengthening of the thumb muscles leads to improved strength.¹⁰⁸ If practical, there is also Level B evidence that training in warm climates and nonswimming water exercises are particularly useful.⁶³ Overall, it is reasonable to prescribe exercise for patients with SMA, KD, and postpolio syndrome within the limitations discussed for ALS earlier.

Pain and Cramps in MND

Cramps are common in patients with various forms of MND. In KD, cramps are early features of the disease and are felt to be related to axonal hyperexcitability and may occur before symptoms of overt motor deficits. Cramps are similarly a common source of discomfort in patients with ALS and are common early in the disease. Mexiletine and ranolazine have been shown to reduced symptoms of muscle cramps.^{35,120,150} Other potential treatments may include massage, physical therapy, magnesium, carbamazepine, phenytoin, verapamil, and gabapentin, though none have been studied in ALS.⁸ Pain is a common feature of ALS in later stages, seen in up to 75% of patients.¹⁷³ Contractures and prolonged immobilization may contribute to this. Treatment therefore should begin with ROM exercises. Physical modalities such as ice and heat can be used. If these are ineffective, acetaminophen or nonsteroidal antiinflammatory medications and antispasticity agents are commonly used. Opioids may be necessary in time. They are effective nearly 75% of the time and have the advantage of relieving air hunger and anxiety.¹⁴⁶ Respiratory depression, loss of airway protection and cough, and constipation need to be monitored and dose-adjusted accordingly.

Bulbar Dysfunction

MNDs with significant bulbar involvement will cause issues with dysarthria and dysphagia and subsequent issues with communication and nutrition. Communication is a major factor in patients' quality of life and can be impaired in any patient with MND and significant bulbar involvement. Early and frequent involvement of speech-language pathologists will help accommodate the resulting barriers to communication through exercises and specialized equipment. Dysphagia in MNDs with bulbar involvement will eventually impact nutrition, and management of nutrition is a major aspect of the overall treatment of patients with severe MNDs. When dysphagia is severe, the swallowing mechanism needs to be bypassed to achieve adequate nutrition. Monitoring of weight and other metrics becomes a necessary part of all follow-up visits with severely affected patients.

Respiratory Insufficiency

Respiratory muscles may be affected by motor neuron loss similar to other muscles. This leads to respiratory insufficiency and, if respiratory function is not supported, eventual respiratory failure.

Respiratory failure is the main source of mortality in patients with MNDs, and the choices and methods of compensating for respiratory insufficiency ultimately can have a significant impact on the lives of patients. Patients who choose tracheostomy and long-term ventilation will generally survive much longer but must consider the eventual possibility of a locked-in, noncommunicative state.

Mood and Cognitive Disorders

Mood disorders are common in MNDs^{185,198,201} and should be screened for and treated because they have a significant impact on the quality of life in patients with MND. Treatment recommendations for patients with MND and mood disorders generally corresponds to the treatment of mood disorders in the general population, with some specific considerations for patients with ALS (see earlier).

Summary

MNDs represent a diverse group of disorders with the common feature of motor neuron loss and associated deficits within the motor system. The goal of this chapter is to provide the general framework to approach patients with suspected MND. The spectrum of disorders reviewed, while not all encompassing, includes the more common of these relatively uncommon disorders. The effects of MND are often devastating, and in most forms of MND, there are generally no cures. Nevertheless, there is a surprising difference that an attentive and educated physician can make on extending and improving the lives of patients with MND, and there is real hope. At the same time as polio is being steadily eradicated, genetic treatments are now available for SMA, and potential treatments for ALS are hopefully on the horizon. In the meantime, a physiatrist's view of the function of the entire patient, ability to coordinate with multiple disciplines, and focus on quality of life are ideal when treating a patient with MND. Proper diagnosis and management of MND have never been more important. A multidisciplinary approach is usually the most effective to address the complexities of diagnosis, prognosis, and management, and physiatrists have key roles in the detailed evaluation, accurate and timely diagnosis, and treatment of patients with MND.

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